Me and my best friend
in serious moment



ACUTE INFLAMMATION

MDP10906 Paraclinical Sciences
2026/2027

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Specific Instructional Objectives

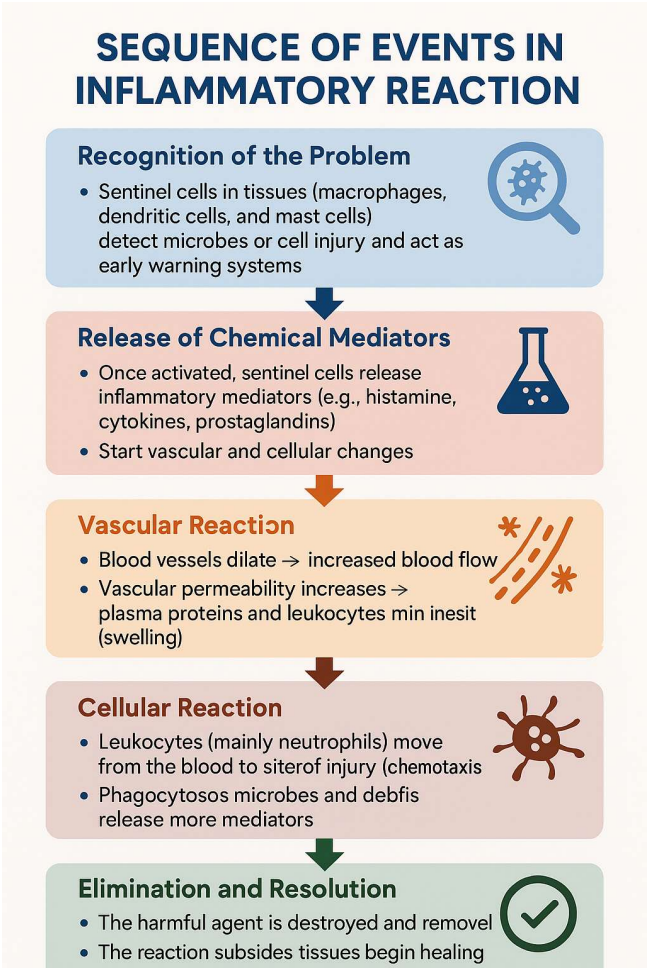
1. Define and state the aim of inflammation
2. List the five cardinal signs of acute inflammation and its pathophysiology.
3. Describe the three components of acute inflammation
    a. Vascular events
    b. Cellular events
    c. Chemical mediators
4. List the causes of acute inflammation.
5. Describe the aetiology and pathogenesis of oedema.
6. Describe the outcomes of acute inflammation.
7. Describe the morphological pattern of acute inflammation.

Overview Of Inflammation: Definitions And General Features

Overview Of Inflammation:

- **Definition:** Inflammation is the protective response of vascular tissues that delivers white blood cells and defense molecules to the site of infection or injury to eliminate the harmful cause.
- **Purpose:** Although it may appear harmful, inflammation is essential for survival — it removes the cause of injury (like microbes or toxins) and the damaged tissue that results.
- **Key Players:**
  - Leukocytes (phagocytes) – destroy pathogens and debris
  - Antibodies and complement proteins – enhance defense
  - Tissue sentinel cells – detect threats early
- **Process Overview:** Inflammation helps transport immune cells and proteins from the blood to the affected tissue, where they become activated to remove harmful substances.
- **Clinical Importance:**
  - Without inflammation:
    - Infections would spread unchecked
    - Wounds would not heal
    - Tissues would remain damaged
- **Terminology:** The suffix “-itis” denotes inflammation (e.g., appendicitis, conjunctivitis, meningitis).

SEQUENCE OF EVENTS IN INFLAMMATORY REACTION



Sequence of events in an inflammatory reaction.

(1) Recognition

- Inflammation begins when the body detects a harmful agent (microbes, toxins, or cell injury).
- Sentinel cells (macrophages, dendritic cells, mast cells) and phagocytes have receptors that recognise these signals.
- When activated, these cells release inflammatory mediators, triggering the next steps of the inflammatory response.

(2) Recruitment

- Leukocytes and plasma proteins move from the blood to the site of injury.
- Because blood flows through all tissues, it can quickly deliver defense cells (neutrophils, monocytes, lymphocytes) and proteins (complement) to the affected area.
- This movement requires changes in blood vessels and release of mediators to allow immune cells and proteins to exit circulation and reach the target site.

Sequence of events in an inflammatory reaction.

(3) Removal

- Phagocytic cells (mainly macrophages and neutrophils) ingest and destroy microbes, dead cells, and debris.
- This clears the cause of inflammation and prepares the area for healingis mainly by phagocytic cells, which ingest and destroy microbes and dead cells.

(4) Regulation

- The inflammatory response must be controlled and stopped once the harmful stimulus is *removed.*
- Anti-inflammatory mediators switch off the reaction to prevent excessive tissue damage.

(5) Repair

- The healing phase begins after inflammation subsides.
- Damaged tissues are restored by:
    - Regeneration of surviving cells, and
    - Formation of scar tissue (fibrosis) to fill remaining defects.

Fundamental properties in inflammation

Components of the inflammatory response.

- The main participants are blood vessels and leukocytes.

1. Blood Vessels

- Dilate to increase blood flow (causing redness and heat).
- Become more permeable to allow plasma proteins to enter the tissue.
- Endothelial cells change to let leukocytes stick and migrate out of circulation.

2. Leukocytes

- Once they enter the tissue, leukocytes become activated.
- They phagocytose and destroy microbes, dead cells, and foreign materials, helping to clear the infection and tissue debris.

Fundamental properties in inflammation

Harmful consequences of inflammation.

- While inflammation is protective, it can also cause tissue damage and disease when excessive or misdirected.

1) Local Damage

- Normal inflammation may cause pain, swelling, and temporary loss of function, but usually heals without lasting injury once the cause is removed.

2) Misdirected or Uncontrolled Inflammation

- Autoimmune diseases: inflammation targets the body's own tissues (e.g. rheumatoid arthritis).
- Allergic reactions: response to harmless substances (e.g. pollen, food, insect stings, drugs).
- Chronic inflammation: underlies diseases like atherosclerosis, lung fibrosis, type 2 diabetes, Alzheimer's disease, and cancer — hence called *"the silent killer."*

3) Clinical Importance

- Anti-inflammatory drugs aim to limit tissue injury while preserving the beneficial effects of inflammation.

Fundamental properties in inflammation

Local and systemic inflammation.

- Local inflammation occurs at a specific site of infection or tissue injury.
  - Example: a cut or abscess.
  - May cause local signs such as redness, swelling, pain, and sometimes mild fever.
- Systemic inflammation happens when the response spreads throughout the body.
  - Seen in severe infections (e.g. sepsis), where inflammation becomes widespread and may cause organ failure.
  - This is called Systemic Inflammatory Response Syndrome (SIRS).

Mediators of inflammation.

- Inflammation is driven by chemical mediators, which are soluble factors released by:
  - Cells (e.g. macrophages, mast cells, endothelial cells), or
  - Plasma proteins (e.g. complement system).
- Triggers: microbes, dead or injured cells, and lack of oxygen (hypoxia).
- These mediators initiate, amplify, and control the inflammatory response — determining its pattern, intensity, and outcome.

Fundamental properties in inflammation

Acute inflammation:

- A rapid and short-lived response to an easily removed cause (e.g. bacteria, fungi, dead cells).
- Develops within minutes to hours and lasts hours to a few days.
- Characterised by:
  - Fluid and plasma protein leakage → oedema
  - Leukocyte migration, mainly neutrophils
- Once the cause is removed, the inflammation subsides and the tissue heals.

Chronic inflammation

- occurs when:
  - Acute inflammation fails to resolve, or
  - It develops de novo (from the start).
- Triggered by persistent or hard-to-eliminate agents (e.g. tuberculosis bacilli, viruses, fungi, or self-antigens).
- Prolonged duration → causes ongoing tissue destruction, scarring, and fibrosis.
- Can coexist with unresolved acute inflammation (e.g. peptic ulcer).

Features of Acute and Chronic Inflammation

Feature	Acute	Chronic
Onset	Fast: minutes or hours	Slow: days
Cellular infiltrate	Mainly neutrophils	Monocytes/macrophages and lymphocytes
Tissue injury, fibrosis	Usually mild and self-limited	Often severe and progressive
Local and systemic signs	Prominent	Less

Historical Highlights

Clinical Features of Inflammation

- The signs of inflammation have been recognized since ancient times (recorded in Egyptian papyrus, ~3000 BC).
- Celsus (1st century AD) described the four cardinal signs of acute inflammation:
    1. Rubor – redness
    2. Tumor – swelling
    3. Calor – heat
    4. Dolor – pain
- Later, Rudolf Virchow (19th century) added a fifth sign:
  5. Functio laesa – loss of function

Historical Insights

- John Hunter (1793): Observed that *inflammation is not a disease*, but a protective response that helps the body heal.
- Elie Metchnikoff (1880s): Discovered phagocytosis — showing that the main purpose of inflammation is to bring phagocytes to the site of injury to engulf microbes and debris.

Cardinal Signs in AI - Pathophysiology

Redness (rubor) & Heat (calor)

- Histamine relaxes arterioles (precapillary sphincters).
- This causes more blood flow into capillaries and venules, making them dilated.
- Results in redness and heat at the inflamed area.
- *(refer notes on reactions of blood vessels)*

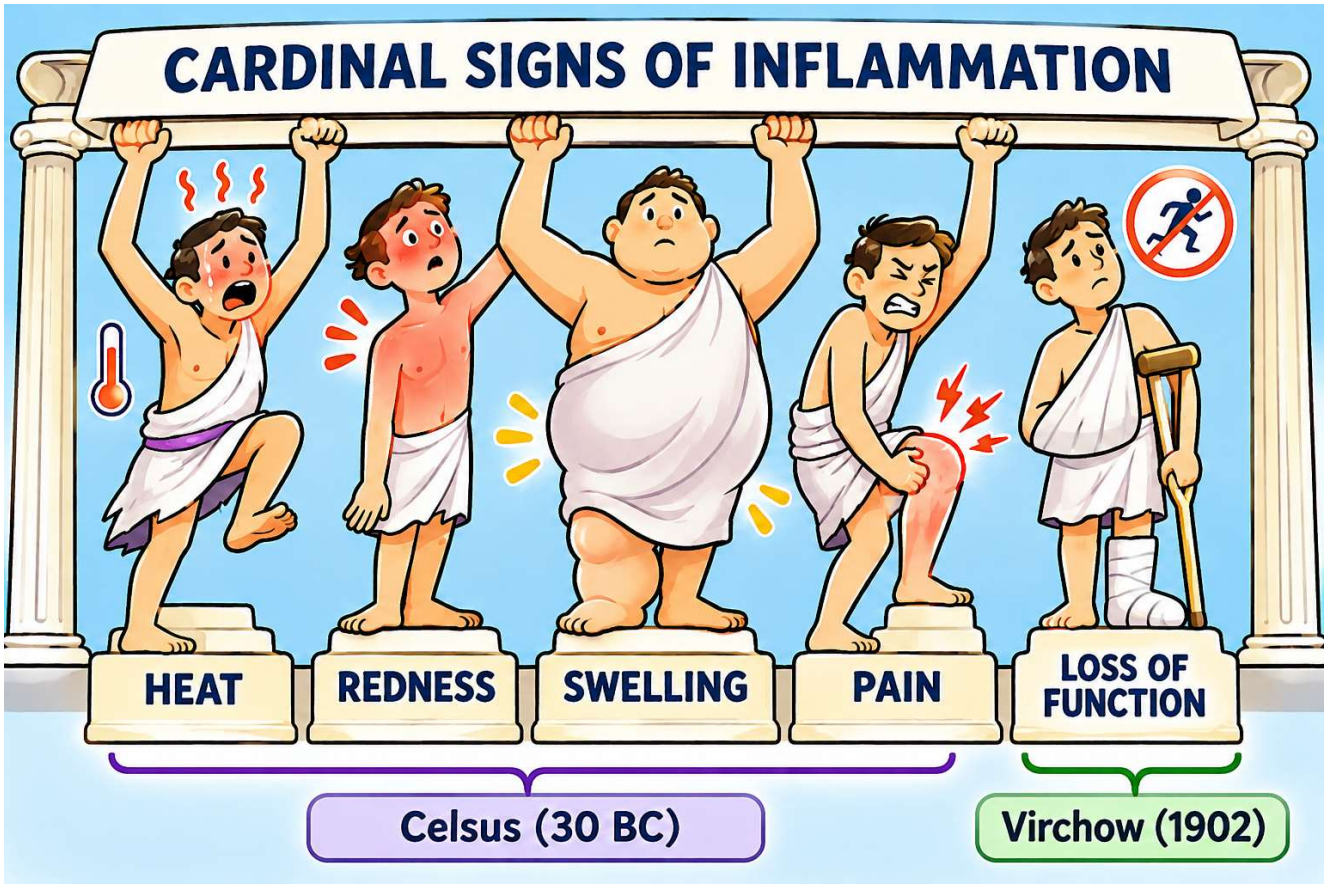
Swelling (Tumour, oedema)

- Histamine increases the permeability of venules, allowing fluid and proteins to leak into surrounding tissues.
- This leakage leads to oedema — the accumulation of excess fluid in the interstitial space.
- *(refer notes on reactions of blood vessels)*

Pain (Dolor)

- Prostaglandin E2 (PGE2) makes nerve endings more sensitive to bradykinin and other pain mediators.
- This increased sensitivity causes the pain (dolor) felt during inflammation.
- *(refer notes on mediators in inflammation)*

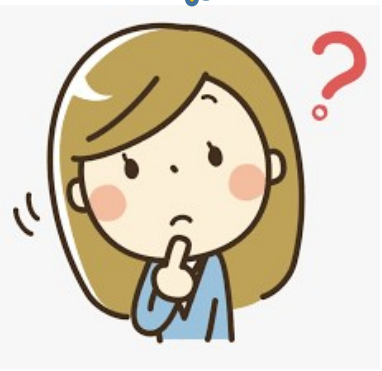
Cardinal signs of acute inflammation



rubor et tumor cum calore et dolore

Functio laesa

Pathophysiology?



Oedema - pathophysiology

Oedema and Fluid Balance

- Definition: Oedema is the accumulation of excess fluid in tissues, while effusion refers to fluid collecting in body cavities (e.g. pleura, pericardium, peritoneum).

Normal Fluid Balance

- Fluid movement across capillaries is controlled by two opposing forces:
  - **Hydrostatic pressure** – pushes fluid out of capillaries into tissues.
  - **Colloid osmotic pressure** (from plasma proteins) – pulls fluid back into capillaries.
- A small amount of fluid normally enters the interstitial space, but it is drained by lymphatic vessels and returned to the bloodstream, keeping tissues dry.

Oedema - pathophysiology

How Oedema Develops

- When this balance is disturbed, more fluid leaves vessels than is drained.
- Causes:
    1. ↑ Hydrostatic pressure (e.g. heart failure)
    2. ↓ Colloid osmotic pressure (e.g. liver or kidney disease)
    3. Blocked lymphatic drainage
- Result:
    - Oedema – fluid in tissues
    - Effusion – fluid in body cavities

Increased Hydrostatic Pressure
Impaired Venous Return
Congestive heart failure
Constrictive pericarditis
Ascites (liver cirrhosis)
Venous obstruction or compression
Thrombosis
External pressure (e.g., mass)
Lower extremity inactivity with prolonged dependency
Arteriolar Dilation
Heat
Neurohumoral dysregulation
Reduced Plasma Osmotic Pressure (Hypoproteinemia)
Protein-losing glomerulopathies (nephrotic syndrome)
Liver cirrhosis
Malnutrition
Protein-losing gastroenteropathy
Lymphatic Obstruction
Inflammatory
Neoplastic
Postsurgical
Postirradiation
Sodium Retention
Excessive salt intake with renal insufficiency
Increased tubular reabsorption of sodium
Renal hypoperfusion
Increased renin-angiotensin-aldosterone secretion
Inflammation
Acute inflammation
Chronic inflammation
Angiogenesis

Oedema - Causes

Types of Oedema and Effusion

- Oedema (tissue fluid) and effusion (fluid in cavities) can be either:
  - Inflammatory, or
  - Non-inflammatory

1. Inflammatory

- Fluid is protein-rich → called exudate.
- Caused by increased vascular permeability during inflammation.
- Contains cells, proteins, and chemical mediators.
- Example: pus or inflammatory fluid in infections.

2. Non-inflammatory

- Fluid is protein-poor → called transudate.
- Due to imbalance in hydrostatic or osmotic pressure, not inflammation.
- Common in heart failure, liver failure, renal disease, or malnutrition.

Oedema - pathophysiology

1. **Increased Hydrostatic Pressure:**
   - Due to impaired venous return (e.g. heart failure, DVT).
   - More fluid pushed out of capillaries into tissues.
2. **Reduced Plasma Osmotic Pressure:**
   - Caused by low albumin from:
     - ↓ Synthesis (e.g. liver disease)
     - ↑ Loss (e.g. nephrotic syndrome)
   - Leads to less fluid drawn back into vessels.
3. **Sodium and Water Retention:**
   - From renal dysfunction or excess salt intake.
   - Causes both:
     - ↑ Hydrostatic pressure (expanded blood volume), and
     - Osmotic pressure (dilution effect).
4. **Lymphatic Obstruction:**
   - Due to trauma, fibrosis, tumors, or infection.
   - Prevents drainage of interstitial fluid, causing lymphedema.
5. **Inflammation:**
   - Leads to increased vascular permeability, allowing protein-rich exudate to escape into tissues.

Inflammation - Causes

- Inflammatory reactions may be triggered by a variety of stimuli:

(1) Infections:

- The most common and important cause of inflammation.
- Caused by bacteria, viruses, fungi, parasites, and microbial toxins.
- Can range from:
  - Mild acute inflammation (heals completely),
  - Severe systemic inflammation (may be fatal), to
  - Chronic inflammation (causes tissue damage).
- The outcome depends on the type of pathogen, the host's immune response, and individual host factors.

(2) Tissue necrosis:

- Any form of cell death can trigger inflammation.
- Causes include ischaemia (e.g. myocardial infarction), trauma, burns, frostbite, radiation, or chemical injury.
- Molecules released from dead cells act as signals that start the inflammatory process.

Inflammation - Causes

(3) Foreign bodies:

- External objects such as splinters, dirt, or sutures can cause inflammation — either directly or through tissue injury or microbial contamination.
- Even internal (endogenous) substances can trigger inflammation when deposited abnormally in tissues:
  - *Urate crystals → gout*
  - *Cholesterol crystals → atherosclerosis*
  - *Lipids → obesity-related inflammation*

(4) *Immune reactions:*

- Occur when the immune system harms the body's own tissues instead of protecting it.
- Includes:
  - *Autoimmune diseases – reaction against self-antigens (e.g. rheumatoid arthritis).*
  - *Allergic reactions – response to harmless environmental substances (e.g. pollen, food).*
  - *Excessive response to microbes – leads to tissue damage.*
- These conditions are chronic and difficult to cure, as the stimuli persist, leading to prolonged inflammation and tissue injury.

Recognition of Microbes & Damaged Cells

- **Recognition of microbial components or substances released from damaged cells is the initiating step in inflammatory reactions.**

(1) Cellular receptors for microbes.

- Cells have receptors that detect microbes in different locations:
  - Plasma membrane → for extracellular microbes
  - Endosomes → for ingested microbes
  - Cytosol → for intracellular microbes

- The most important are Toll-like Receptors (TLRs).
  - Found on epithelial cells, macrophages, dendritic cells, and leukocytes.
  - When activated, they trigger the release of inflammatory mediators (e.g. cytokines, chemokines).

Recognition of Microbes & Damaged Cells

(2) Sensors of cell damage.

- All cells have cytosolic receptors that detect cell injury or stress.
- The main group is the *NOD-like receptors (NLRs).*

What they detect:

- Uric acid – from DNA breakdown
- ATP – released from damaged mitochondria
- ↓ Intracellular K+ – due to membrane injury
- Cytoplasmic DNA – normally confined to the nucleus

What happens next:

- These signals activate the inflammasome, a protein complex inside the cell.
- The inflammasome triggers the release of Interleukin-1 (IL-1), a cytokine that recruits leukocytes and starts inflammation.

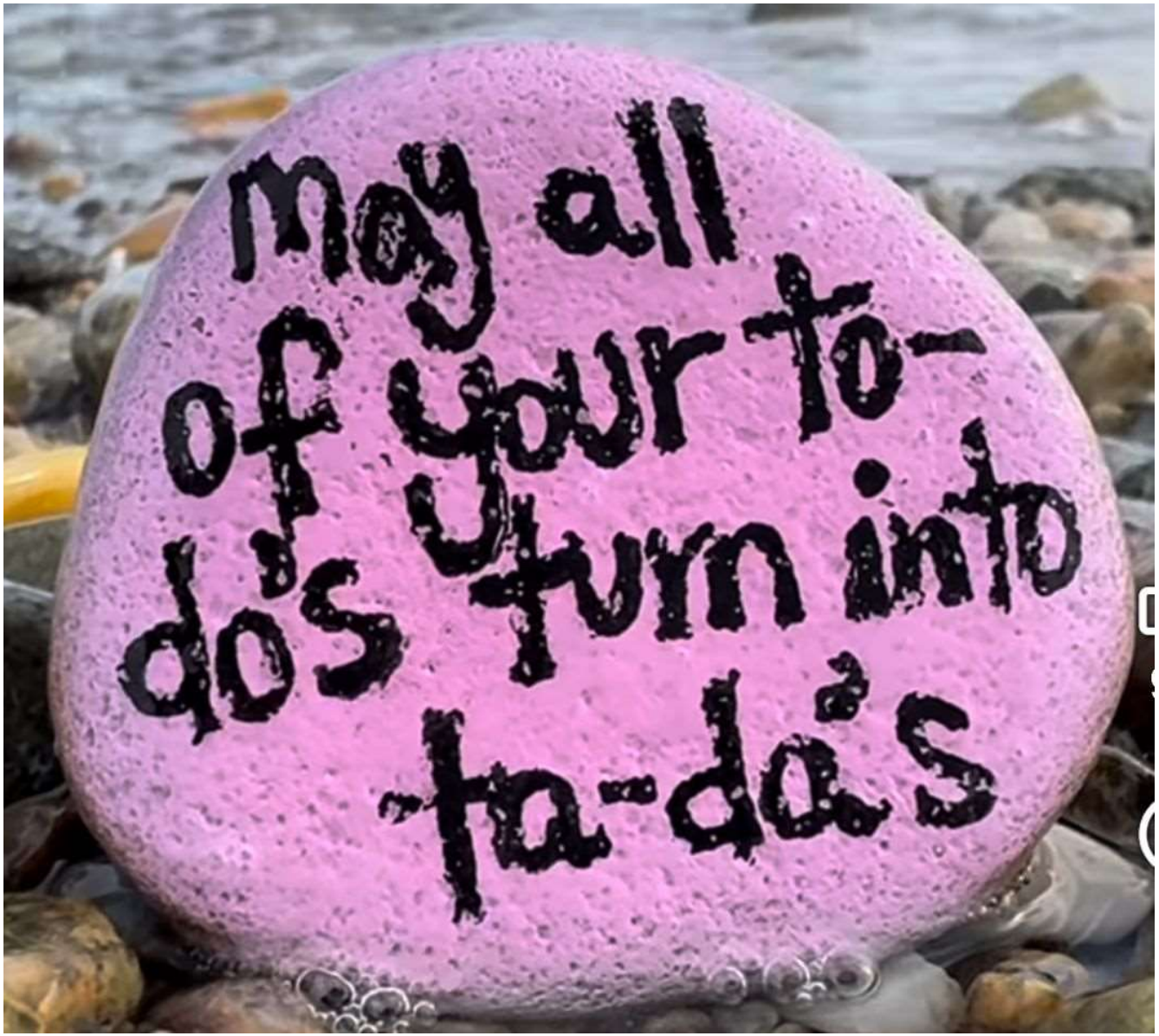
Recognition of Microbes & Damaged Cells

(3) Other cellular receptors involved in inflammation.

- Leukocytes have receptors that recognise microbes coated with:
  - Antibodies, or
  - Complement proteins.
- This coating process is called opsonisation.
- These receptors help phagocytose (ingest) and destroy microbes, while also amplifying inflammation.

(4) Circulating proteins.

- Complement system – reacts against microbes and generates inflammatory mediators.
- Mannose-binding lectin (MBL) – recognises sugars on microbes, helps with phagocytosis and activates complement.
- Collectins – other plasma proteins that bind and neutralise microbes.


may all
of your to-
do's turn into
-ta-da's

ACUTE INFLAMMATION

a) Reactions of Blood Vessels
b) Leukocyte Recruitment to Sites of Inflammation
c) Phagocytosis and Clearance of the Offending Agent
d) Termination of the Acute Inflammatory Response
e) Mediators of Inflammation
f) Morphologic Patterns of Acute Inflammation
g) Outcomes of Acute Inflammation
h) Summary of Acute Inflammation

Acute Inflammation

Major Components of Acute Inflammation

Acute inflammation has three key components:

1. **Dilation of small vessels**
   - Increases blood flow to the affected area.
   - Causes redness (rubor) and heat (calor).
2. **Increased vascular permeability**
   - Plasma proteins and leukocytes leak out of blood vessels into tissues.
   - Leads to oedema (swelling).
3. **Emigration of leukocytes**
   - White blood cells leave the circulation, accumulate at the injury site, and become activated to destroy the harmful agent.

Initiation of the Response

- When a microbe or dead cell is detected → resident phagocytes (macrophages, dendritic cells, mast cells) attempt to eliminate it.
- These cells release cytokines, lipid mediators, and other signals, which:
  - Act on nearby blood vessels,
  - Cause fluid and leukocyte movement into tissues.

Acute inflammation

Vascular events

(1) Arteriolar vasodilation
→ ↑ blood flow → redness & heat

(Brief vasoconstriction → neurogenic reflex (few seconds).

(2) Increased Vascular Permeability

Venules become leaky → due to histamine and other mediators

Plasma proteins & leukocytes escape into tissues

Causes oedema (swelling) at the site of injury

Cellular events

(3) Recruitment of Leukocytes

Leukocytes exit microcirculation (emigration)

Accumulate at the site of injury

Activate to destroy microbes and clear debris

Acute Inflammation: (a) Reactions of Blood Vessels (Vascular events)

Acute inflammation involves changes in blood flow and vessel permeability, allowing plasma proteins and leukocytes to move from the circulation to the site of infection or injury.

(a) Exudate

- Fluid with high protein content and cellular debris.
- Indicates an inflammatory process with increased vascular permeability.

(b) Transudate

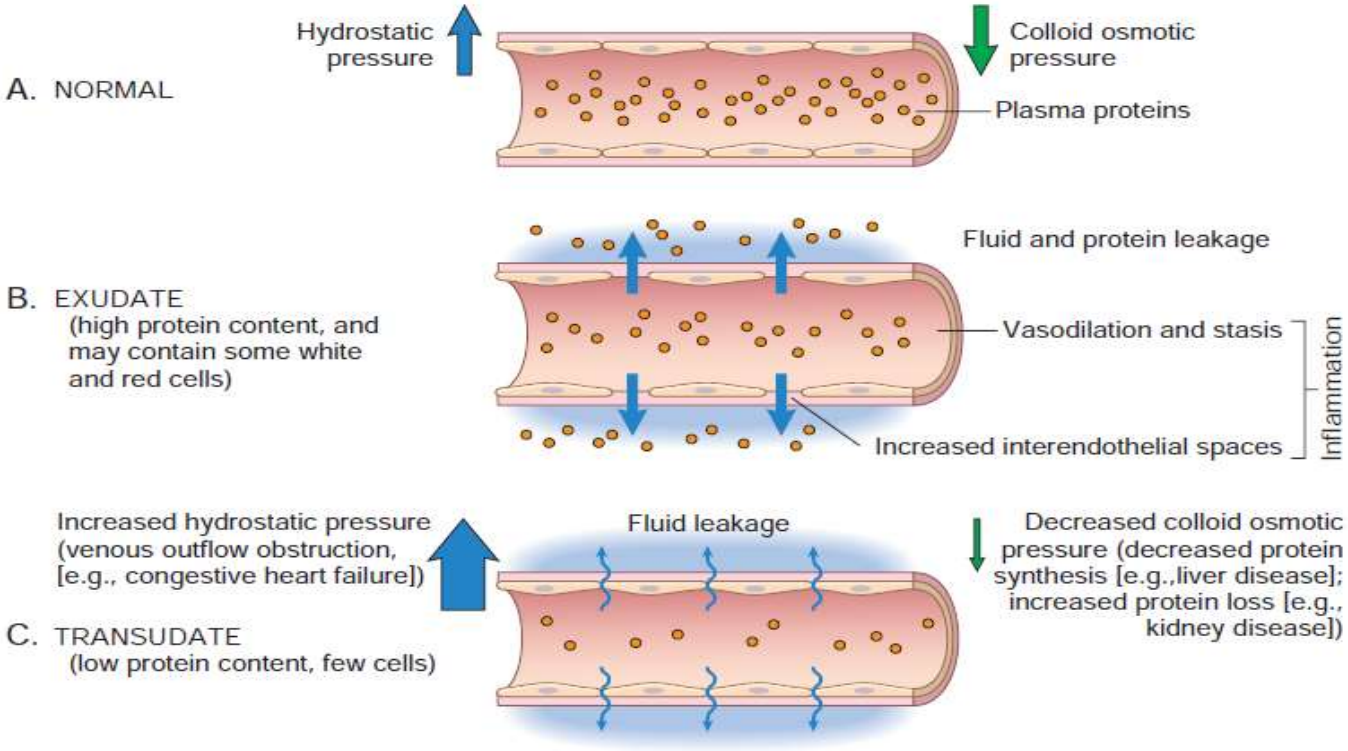
- Fluid with low protein content (mostly albumin) and few or no cells.
- Caused by osmotic or hydrostatic imbalance, not inflammation.

(c) Oedema

- Excess fluid in tissues or body cavities.
- Can be either exudate or transudate.

(d) Pus (Purulent Exudate)

- An inflammatory exudate rich in neutrophils, dead cells, and often microbes.



Formation of Exudates &Transudates

Fluid Movement Across Capillaries

(A) Normal pressures:

- Hydrostatic pressure: ~32 mm Hg (arterial end) → ~12 mm Hg (venous end)
- Colloid osmotic pressure: ~25 mm Hg

- Because these forces balance each other, there is almost no net fluid movement under normal conditions.

In Pathologic Conditions

(B) Exudate → formed in inflammation due to increased vascular permeability and wider endothelial gaps.

(C) Transudate → formed when hydrostatic pressure increases (e.g. heart failure) or osmotic pressure decreases (e.g. low plasma proteins).

Vascular Event In Acute Inflammation

Immediate transient vasoconstriction of arterioles

↓ Histamine, prostaglandins, kinins and NO

Persistent progressive vasodilatation of arterioles

↓

Opening of new capillary beds (local heat and redness)

↓ Histamine

Increased vascular permeability: Endothelial contraction , endothelial cell injury

hallmark of acute inflammation

↓

Exudation of fluid into the extracellular space: Oedema

↓

Stasis

↓

Leucocytic margination or peripheral orientation of WBCs along the endothelial surface (neutrophils stick to the endothelium and migrate through vascular wall into the interstitial tissue. RBCs are smaller, move faster and are oriented centrally)

Reactions of blood vessel: *(i) Changes in Vascular Flow and Caliber*

- **Vasodilation** is one of the earliest events in acute inflammation.
  - Triggered by *histamine* and other mediators acting on vascular smooth muscle.
  - Begins in arterioles and then opens new capillary beds, increasing blood flow.
  - Results in heat and redness (erythema) at the inflamed site

- **Increased vascular permeability** follows, allowing protein-rich fluid to leak into tissues.
  - Fluid loss + vessel dilation → slower blood flow, red cell concentration, and increased viscosity.
  - This causes vascular congestion and stasis (sluggish circulation).

- During **stasis**, neutrophils move to the vessel margins, where endothelial cells express more adhesion molecules.
  - Leukocytes adhere to the endothelium and migrate into tissues to begin the inflammatory response.

Reactions of blood vessel:

(ii) Increased Vascular Permeability (Vascular Leakage)

- Several mechanisms are responsible for the increased permeability of postcapillary venules, a ***hallmark*** of acute inflammation

(a) Contraction of endothelial cells:

- The most common cause of vascular leakage.
- Mediated by histamine, bradykinin, and leukotrienes.
- Causes interendothelial gaps → allowing plasma and proteins to escape.
- Known as the **"immediate transient response"** — occurs within minutes and lasts 15–30 minutes.

Delayed Prolonged Leakage

- Seen in mild injuries (e.g. burns, UV exposure, radiation, bacterial toxins).
- Leakage begins after 2–12 hours and can last hours to days.
- Example: sunburn — classic case of delayed vascular leakage.
- Both immediate and delayed responses may occur in sequence.

Reactions of blood vessel:

(ii) Increased Vascular Permeability (Vascular Leakage)

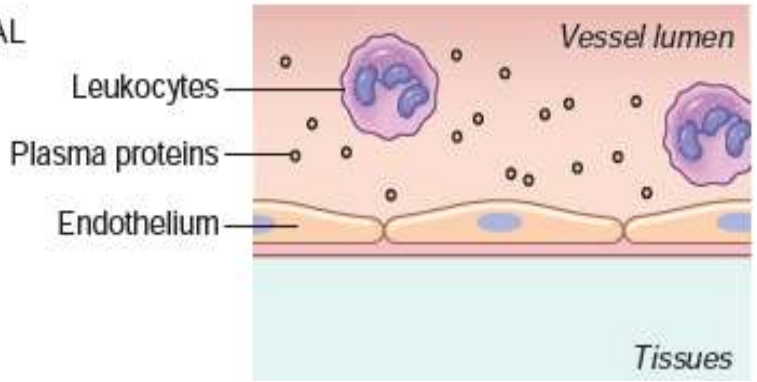
(b) Endothelial injury:

- Caused by direct damage to endothelial cells leading to cell necrosis and detachment.
- Seen in severe physical injuries (e.g. thermal burns) or microbial toxin damage.
- Neutrophils adhering to endothelium can also cause secondary injury, worsening the leak.
- Leakage begins immediately after injury and lasts for hours, until vessels are thrombosed or repaired.

- Although described separately, multiple mechanisms often act together.
- *Example*: In a thermal burn, leakage results from chemical mediators, direct injury, and neutrophil-induced damage — leading to major fluid loss that can be life-threatening.

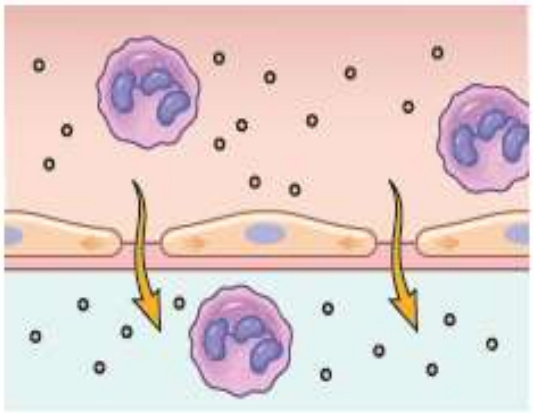
Principal mechanisms of increased vascular permeability in inflammation and their features and underlying causes.

A. NORMAL



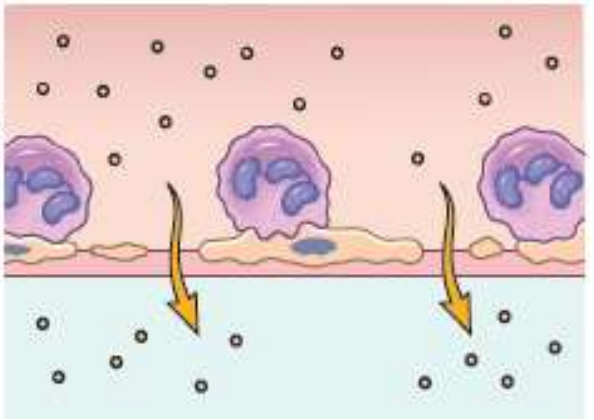
B. RETRACTION OF ENDOTHELIAL CELLS

- Induced by histamine, other mediators
- Rapid and short-lived (minutes)



C. ENDOTHELIAL INJURY

- Caused by thermal burns, some microbial toxins
- Rapid; may be long-lived (hours to days)



Reactions of blood vessel:

(iii) Responses of Lymphatic Vessels and Lymph Nodes

Role of Lymphatic Vessels in Inflammation

- Lymphatic vessels help drain excess fluid, proteins, and cells from inflamed tissues.
- In normal conditions, they drain the small amount of fluid that escapes from capillaries.
- During inflammation, lymph flow increases to remove oedema fluid, leukocytes, cell debris, and microbes.
- Lymphatic vessels proliferate to handle this increased load.

Lymph Node Response

- Draining lymph nodes become enlarged due to:
  - Hyperplasia of lymphoid follicles, and
  - Increased lymphocytes and macrophages.
- This reaction is called reactive (inflammatory) lymphadenitis.


lymphangitis

Clinical Significance

- Red streaks near a wound = inflamed lymphatic channels → lymphangitis.
- Painful, enlarged lymph nodes = lymphadenitis.


lymphadenitis

Acute Inflammation:

(b)Leukocyte Recruitment to Sites of Inflammation (Cellular events)

Leukocyte Recruitment in Inflammation

- After vascular changes, there is a rapid influx of leukocytes into the inflamed tissue
- Their main role is to eliminate the offending agents.

Key Cells

- Neutrophils and macrophages are the main phagocytic cells.
- They ingest and destroy microbes, necrotic tissue, and foreign material.
- Macrophages also release growth factors that promote tissue repair.

Collateral Damage

- When activated, leukocytes release enzymes and reactive substances that help kill microbes.
- However, these same products can also injure normal tissues, causing prolonged inflammation.

Acute Inflammation:

(b)Leukocyte Recruitment to Sites of Inflammation (Cellular events)

Leukocyte Journey from Blood to Tissue

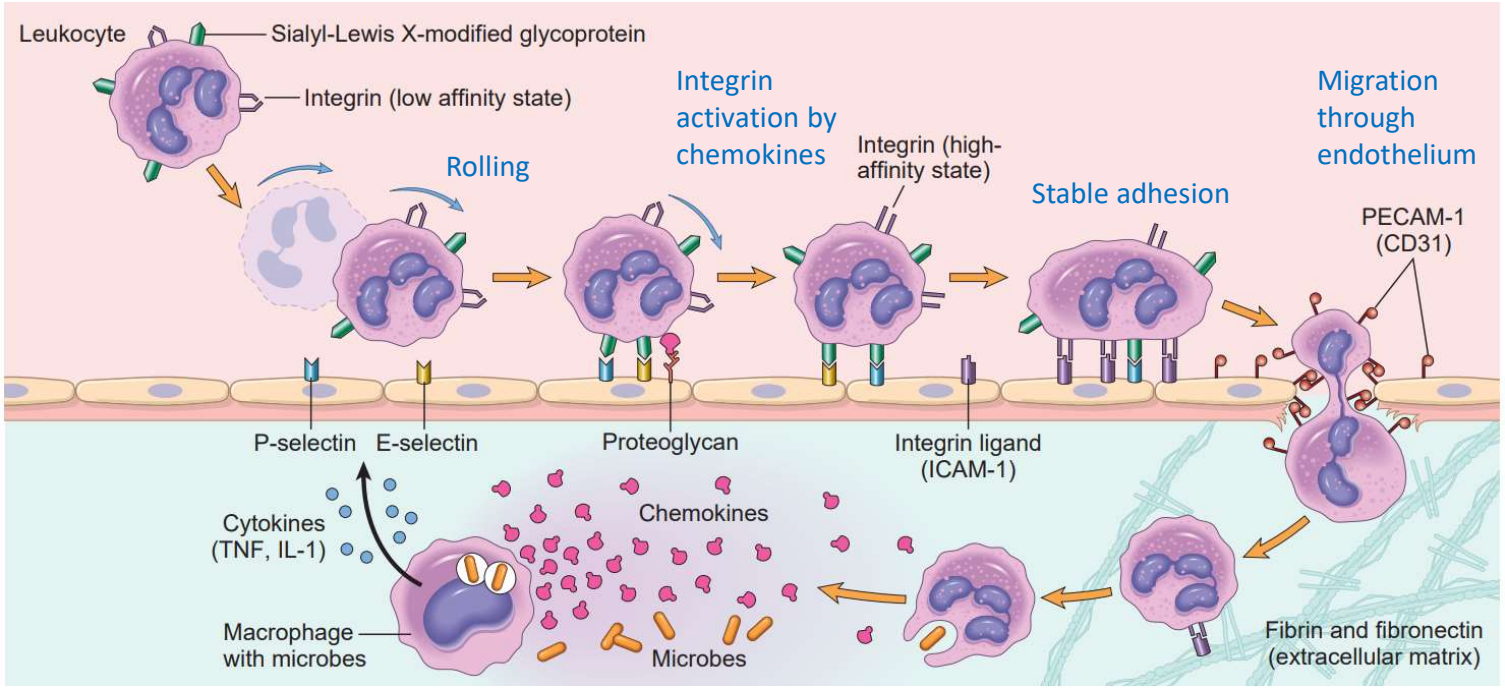
- The movement of leukocytes from the blood vessel lumen into the tissue is guided by adhesion molecules and chemokines**.**

Phases of Leukocyte Migration

Cellular event

1. **In the Vessel Lumen:**
   - Margination, rolling, and adhesion to the endothelium.
   - Normally, endothelium does not bind leukocytes, but during inflammation it becomes activated and expresses adhesion molecules, allowing leukocytes to attach.
2. **Transmigration (Diapedesis):**
   - Leukocytes move across the endothelium and vessel wall into the surrounding tissue.
3. **Migration in Tissue (Chemotaxis):**
   - Leukocytes move toward the site of injury or infection, guided by chemotactic signals.

The multistep process of leukocyte migration through blood vessels, for neutrophils



Cellular Event

Normal Blood Flow:

- RBC flow in the center of the vessel,
- WBC move toward the vessel wall because they are heavier and slower.

In inflammation, leukocytes:

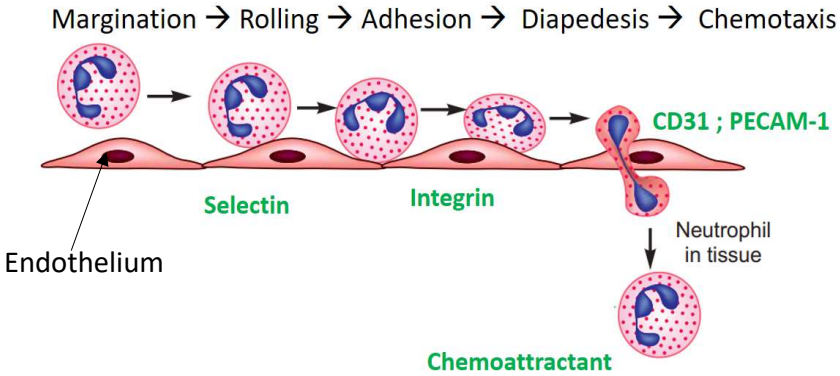
- Roll along the endothelium,
- Activate and firmly adhere,
- Cross the endothelium (diapedesis),
- Pierce the basement membrane
- Migrate toward the site of injury following chemotactic signals.

Acute Inflammation:

(b)Leukocyte Recruitment to Sites of Inflammation (Cellular events)

Leukocyte Adhesion to Endothelium

- In normal blood flow, RBCs move in the center of the vessel, pushing WBCs toward the vessel wall.
- During inflammation, vasodilation causes slower blood flow (stasis), allowing more leukocytes to move to the vessel periphery.
- This redistribution of leukocytes toward the endothelium is called margination.
- The slowed leukocytes detect signals from the activated endothelium, begin to roll, and then firmly adhere through adhesion molecules — like *pebbles* on a streambed that stay in place despite the flow.



Acute Inflammation:
(b)Leukocyte Recruitment to Sites of Inflammation

Leukocyte Adhesion Molecules

- Leukocyte attachment to the endothelium is controlled by adhesion molecules, whose expression increases in response to cytokines during inflammation.
- The two main families involved are selectins and integrins.
- These molecules are found on both leukocytes and endothelial cells and help guide rolling, adhesion, and migration.

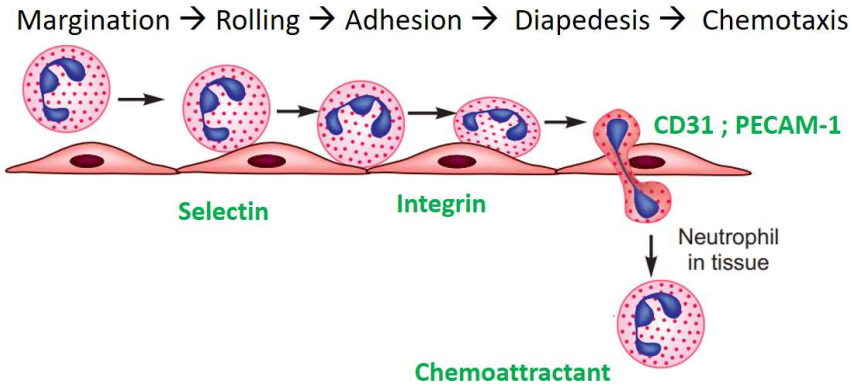
(1) Selectins

- Responsible for the initial rolling of leukocytes along the endothelium.
- Types of selectins:
  - L-selectin → on leukocytes
  - E-selectin → on endothelial cells
  - P-selectin → on platelets and endothelium
- Cytokines (released during infection or injury) increase the expression of selectins and their ligands to promote leukocyte–endothelial interaction.

Acute Inflammation:
(b)Leukocyte Recruitment to Sites of Inflammation

(2) Integrins

- After rolling, leukocytes slow down and get ready for firm attachment.
- Integrins on leukocyte surfaces bind strongly to the endothelium.
- Cytokines (TNF and IL-1) activate endothelial cells to express integrin ligands.
- This binding creates firm adhesion, anchoring leukocytes before they migrate into tissues.



Acute Inflammation:
(c) Leukocyte Migration Through Endothelium

Transmigration (Diapedesis)

- After firm adhesion, leukocytes cross the endothelium — a process called transmigration or diapedesis.
- This mainly occurs in postcapillary venules.
- Chemokines guide the adherent leukocytes to move through gaps between endothelial cells toward the site of injury or infection (chemotactic gradient).

- **Key Molecules**
  - CD31 / PECAM-1 (Platelet Endothelial Cell Adhesion Molecule) – helps leukocytes pass through endothelial junctions.
  - Collagenases – secreted by leukocytes to pierce the basement membrane and enter the tissue.
  - The basement membrane then reseals once the leukocytes have crossed.

Acute Inflammation: (d) Chemotaxis of Leukocytes

Chemotaxis

- After leaving the bloodstream, leukocytes migrate toward the site of injury through a process called chemotaxis — movement along a chemical gradient.

Types of Chemoattractants

- Exogenous:
  - Bacterial products (strong chemoattractants).
- Endogenous:
  1. Cytokines – especially chemokines (e.g. IL-8)
  2. Complement component C5a
  3. Arachidonic acid metabolite LTB_4 (Leukotriene B4)

Acute Inflammation: (d) Chemotaxis of Leukocytes

Leukocyte Sequence

- **Neutrophils** → dominate during the first 6–24 hours.
- **Monocytes/Macrophages** → replace them in 24–48 hours.
- Type of infiltrating cell may vary depending on the stimulus and stage of inflammation.

Early Dominance of Neutrophils

- Neutrophils are the first leukocytes to arrive at the site of inflammation.
- *Reasons for their early appearance:*
  - They are more numerous in the blood.
  - They respond faster to chemokines.
  - They bind more strongly to P-selectin and E-selectin on activated endothelium.

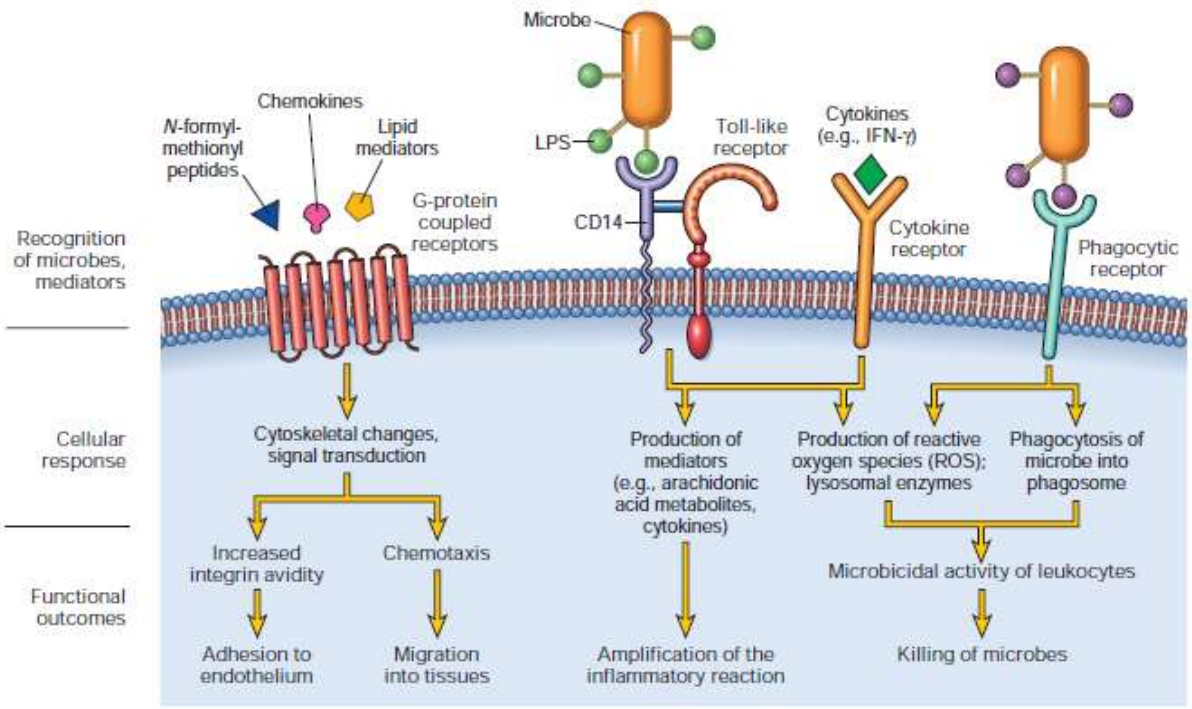
Lifespan and Replacement

- After entering tissues, neutrophils are short-lived and undergo apoptosis within a few days.
- Monocytes survive longer and can proliferate in tissues, becoming the dominant cells in prolonged inflammation.

Leukocyte Activation

- Once at the site of infection or injury, neutrophils and monocytes must be activated.
- Activation occurs through pattern-recognition receptors such as Toll-like receptors (TLRs).
- These receptors trigger phagocytosis and destruction of microbes or dead cells.

Leukocyte activation



Leukocyte Surface Receptors

1. Recognition: Leukocyte receptors detect microbes, dead cells, or signals.
2. Activation: Binding triggers cell activation and defense responses.
3. *Examples:*
   - LPS → binds LPS-binding protein → activates leukocytes
   - IFN-γ → activates macrophages → enhances microbe killing



(c) Phagocytosis and Clearance of the Offending Agent

i. Phagocytosis
ii. Intracellular Destruction of Microbes and Debris
iii. Neutrophil Extracellular Traps
iv. Leukocyte-Mediated Tissue Injury
v. Other Functional Responses of Activated Leukocytes

Phagocytosis and Clearance of the Offending Agent: *(i) Phagocytosis*

Phagocytosis and Leukocyte Activation

- The two main phagocytic cells: Neutrophils and Macrophages.
- Recognition of microbes or dead cells triggers leukocyte activation.
- Activation involves cell signaling, leading to:
  - ↑ Cytosolic Ca^{2+}
  - Activation of enzymes (Protein kinase C, Phospholipase A_2)

Main Functional Responses

- Phagocytosis
- Intracellular killing of microbes

Steps of Phagocytosis

1. Recognition & Attachment – Binding of microbe or particle to leukocyte receptors.
2. Engulfment – Formation of a phagocytic vacuole (phagosome).
3. Killing & Degradation – Destruction of the ingested material by enzymes and reactive oxygen species (ROS).

Phagocytosis and Clearance of the Offending Agent: *(i) Phagocytosis*

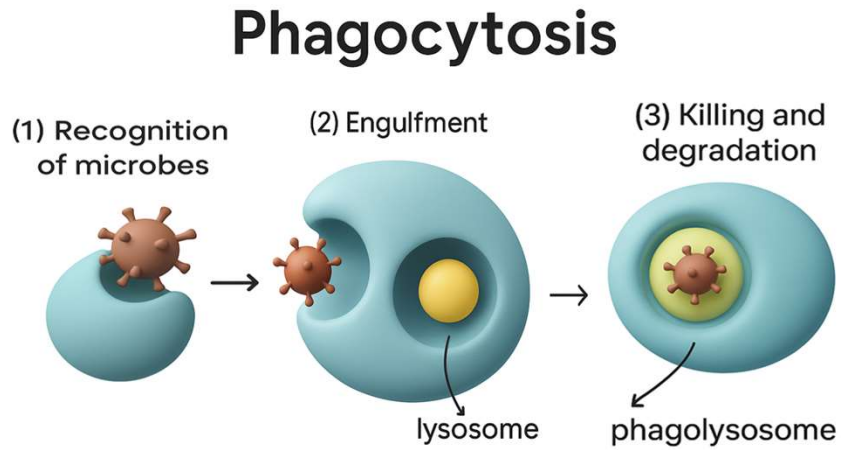
(1) Recognition and attachment

- Phagocytosis begins when phagocytes recognise and bind microbes through specific surface receptors.
- **Phagocytic Receptors**
  - Mannose Receptors – Recognise mannose and fucose residues on microbial cell walls, not on host cells.
  - Scavenger Receptors – Bind modified LDL and a wide range of microbes for uptake.
- **Opsonins**
  - Opsonins are molecules that coat microbes to enhance phagocytosis.
  - Phagocytes have high-affinity receptors for these opsonins.
  - Major opsonins:
    - IgG antibodies
    - C3b (complement fragment)
    - Mannose-binding lectin and collectins

Phagocytosis and Clearance of the Offending Agent: *(i) Phagocytosis*

(2) Engulfment.

- Once a particle binds to phagocyte receptors, the cytoplasm extends around it.
- The plasma membrane encloses the particle, forming a phagosome (internal vesicle).
- The phagosome fuses with a lysosome to form a phagolysosome, where digestion occurs.
- Some lysosomal contents may also be released outside the cell, contributing to tissue damage in inflammation.



Phagocytosis and Clearance of the Offending Agent:

(ii) Intracellular Destruction of Microbes and Debris

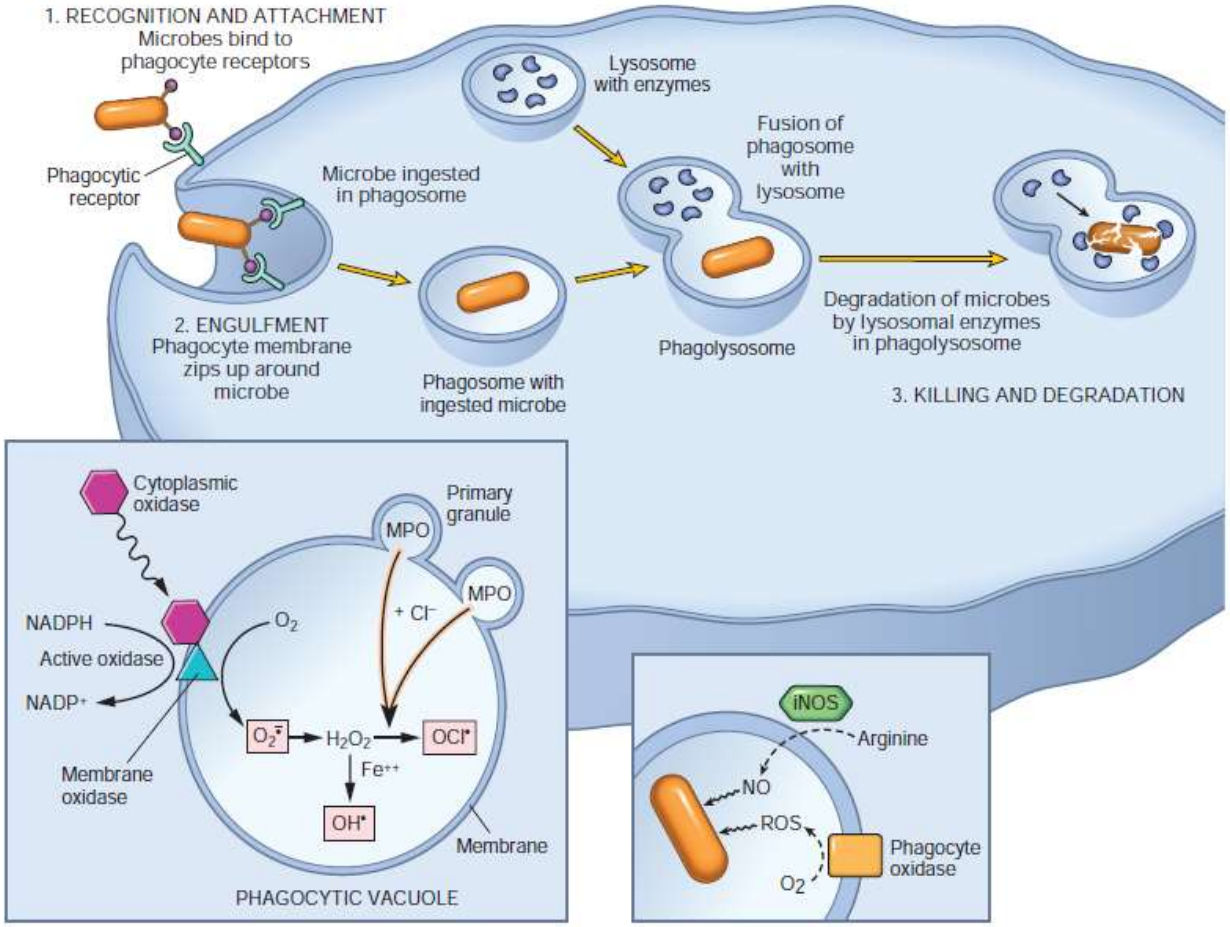
(3) Killing and Degradation - Phagocytosis

- Final step in eliminating infectious agents and necrotic cells.
- Occurs inside activated neutrophils and macrophages after phagocytosis.

- **Mechanisms of killing:**
  - **Oxygen-dependent:**
    - Myeloperoxidase (MPO) system.
    - Oxygen-derived free radicals (ROS).
  - **Oxygen-independent:**
    - Lysosomal enzymes.
    - Reactive nitrogen species (mainly nitric oxide, NO).

- All killing mechanisms are contained within lysosomes, where the phagocytosed materials are delivered.
- This segregation protects the cell's cytoplasm and nucleus from damage while ensuring effective microbial killing and debris degradation.

Phagocytosis and intracellular destruction of microbes.



Phagocytosis and Killing

- **Engulfment:** Microbe binds to receptors → cell membrane wraps around it → forms a phagosome.
- **Fusion:** Phagosome + Lysosome → Phagolysosome.
- **Killing**: Inside the phagolysosome:
  - Enzymes digest microbes.
  - ROS and NO kill microbes.
  - MPO enzyme forms hypochlorite (HOCl).
  - Hydroxyl (·OH) and peroxynitrite (OONO·) also destroy microbes.
- **Release**: Some enzymes may leak into tissues.

Phagocytosis and Clearance of the Offending Agent:

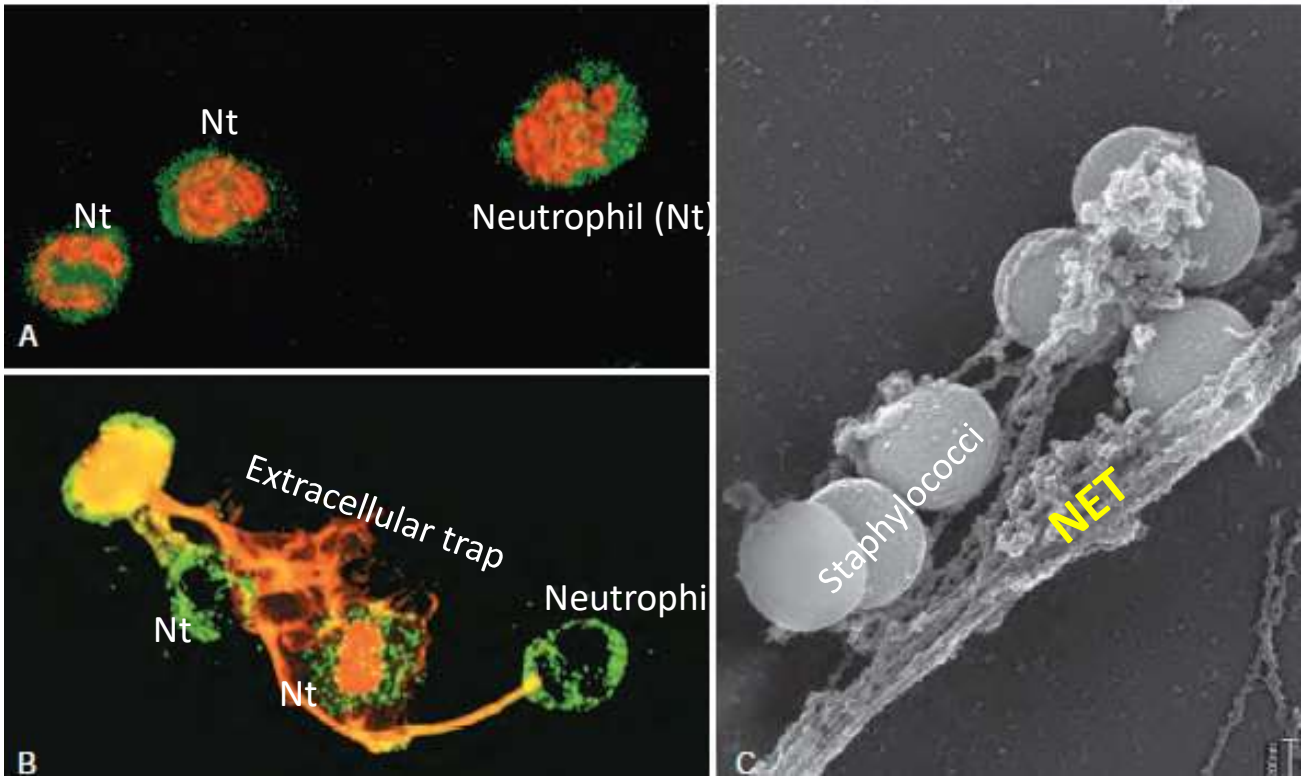
(iii) Neutrophil Extracellular Traps

(3) Killing and Degradation — Neutrophil Extracellular Traps (NETs)

- NETs are web-like structures made of DNA (chromatin) and granule proteins released by neutrophils.
- Their role is to trap and kill microbes (mainly bacteria and fungi) and prevent infection spread.

- **Formation of NETs**
  - Triggered by infectious agents and inflammatory mediators (chemokines, cytokines like interferons, complement, and ROS).
  - ROS activate enzymes that convert arginine → citrulline, causing chromatin to unwind.
  - MPO and elastase move into the nucleus, further loosening DNA.
  - The nuclear membrane ruptures, and chromatin + antimicrobial proteins are released into the extracellular space as a NET.
  - During this process, the neutrophil dies (a type of cell death distinct from apoptosis).

(iii) Neutrophil extracellular traps (NETs)



A. Healthy neutrophils with nuclei stained red and cytoplasm stained green.
B. Release of nuclear material from neutrophils (note that two have lost their nuclei), forming extracellular traps.
C. Electron micrograph of bacteria (staphylococci) trapped in NETs.

(*From Brinkmann V, Zychlinsky A: Beneficial suicide: why neutrophils die to make NETs, Nat Rev Microbiol 5:577, 2007,*

Phagocytosis and Clearance of the Offending Agent:

(iv) Leukocyte-Mediated Tissue Injury

(4) Leukocyte-Mediated Tissue Injury

- Leukocytes can also damage normal tissues while defending the body.
- This injury occurs when their defense mechanisms are activated inappropriately or excessively.

- **Situations causing tissue injury**
  1. Normal defense reactions: Collateral damage to nearby healthy tissue during infection (e.g. tuberculosis, viral infections).
  2. Autoimmune diseases: Inflammation is misdirected against self-tissues.
  3. Allergic reactions: Excessive response to normally harmless substances (e.g. asthma, allergies).

- **Mechanism**
  - The same weapons (enzymes, ROS, cytokines) used to kill microbes can also injure host cells.
  - Once activated, leukocytes cannot distinguish between microbes and normal tissues.

Phagocytosis and Clearance of the Offending Agent:
(iv) Leukocyte-Mediated Tissue Injury

(4) Leukocyte-Mediated Tissue Injury

- **Leukocyte-Mediated Tissue Injury**
  - During activation and phagocytosis, neutrophils and macrophages generate microbicidal substances (ROS, NO, lysosomal enzymes).
  - These are meant to kill microbes inside the phagolysosome, but some are also released outside the cell.
- **How Tissue Injury Occurs**
  - Extracellular release of ROS, NO, and enzymes → can damage host tissues.
  - If not well controlled, leukocytes themselves become the cause of injury.
  - This contributes to many acute and chronic inflammatory diseases.

- **Mechanisms of Extracellular Release**

1. **Controlled Secretion**
   - Activated leukocytes can intentionally release some granule contents into tissues.
   - This is a normal response but can cause collateral damage.
2. **Frustrated Phagocytosis**
   - Occurs when leukocytes try to ingest something too large or fixed, such as:
     - Immune complexes on the glomerular basement membrane
   - Since they cannot engulf the target, they release lysosomal enzymes into the extracellular space, causing tissue injury.
3. **Phagolysosome Membrane Damage**
   - Some ingested materials (e.g. urate crystals) can rupture the phagolysosome. (e.g Gout)
   - This releases lysosomal enzymes into the cytoplasm and extracellular space → worsening inflammation and tissue damage.

Phagocytosis and Clearance of the Offending Agent:

(iv) Leukocyte-Mediated Tissue Injury

Conditions with defective leukocyte function:

Leukocyte defects may be genetic or acquired, and they impair chemotaxis, adhesion, phagocytosis, or microbial killing

A. Genetic Disorders

1. **Chediak–Higashi Syndrome**
   - Defect in lysosomal granules
   - Lysosomes cannot fuse with phagosomes → no phagolysosome formation
   - Results in recurrent bacterial infections
2. **Chronic Granulomatous Disease (CGD)**
   - X-linked or autosomal recessive
   - Defect in NADPH oxidase → no ROS generation
   - Leukocytes cannot kill microbes effectively → forms granulomas
3. **Myeloperoxidase (MPO) Deficiency**
   - Loss of MPO–H_2O_2 system
   - Impaired formation of hypochlorite (HOCl)
   - Mild susceptibility to fungal infections (e.g. Candida)

B. Acquired Disorders

1. Defective Chemotaxis

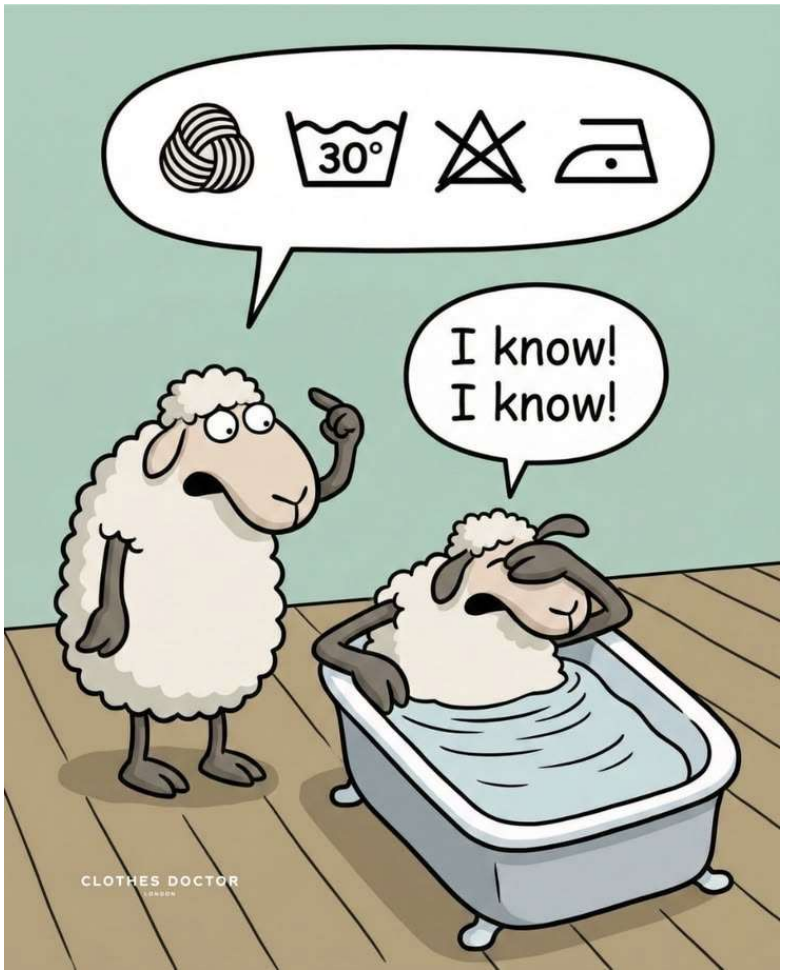
- Seen in: Thermal injury, Diabetes, Malignancy, Sepsis, Immunodeficiency states

2. Defective Adhesion

- Occurs in: Diabetes, Haemodialysis patients

3. Defective Phagocytosis / Microbicidal Activity

- Seen in: Leukaemia, Anaemia, Sepsis, Diabetes, Neonates, Malnutrition


30°
I know!
I know!
CLOTHES DOCTOR

(d) Termination of the Acute Inflammatory Response

(d) Termination of the Acute Inflammatory Response

- Inflammation is essential for host defense but can damage tissues, so it must be tightly regulated.

Why Inflammation Declines

1. **Removal of the stimulus**
   - Mediators are only produced while the offending agent is present.
   - Once it is eliminated, mediator production stops.
2. **Short half-life of mediators**
   - Chemical mediators are rapidly degraded, so their effects fade quickly.
3. **Short lifespan of neutrophils**
   - Neutrophils survive only hours in tissues and undergo apoptosis, reducing inflammation.
4. **Active stop signals**
   - The inflammatory process itself generates anti-inflammatory signals (e.g., lipoxins, IL-10, TGF-β) that actively switch off the reaction.

See



problems



as opportunities



(e) Mediators of Inflammation

i. Vasoactive Amines: Histamine and Serotonin
ii. Arachidonic Acid Metabolites
iii. Cytokines and Chemokines
iv. Complement System
v. Other Mediators of Inflammation

(e) Mediators of Inflammation

- Inflammatory mediators are substances that start, amplify, and regulate inflammation.
- **The key mediators for acute inflammation are:**
  - Vasoactive amines → histamine, serotonin
  - Lipid mediators → prostaglandins, leukotrienes
  - Cytokines & chemokines → TNF, IL-1, IL-6, IL-8
  - Complement fragments → C3a, C5a ("anaphylatoxins")

Where do mediators come from?

A. Cell-derived mediators

- Stored in granules → released rapidly
  - Example: Histamine (mast cells)
- Synthesised on demand → slower release
  - Examples: Prostaglandins, leukotrienes, cytokines
- Major cell sources:
  - Macrophages
  - Dendritic cells,
  - Mast cells
  - (also platelets, neutrophils, endothelial cells)
- Different mediators have short half-lives, so inflammation is limited unless the stimulus persists.

B. **Plasma-derived mediators**

- Made by the liver
- Circulate as inactive precursors
- Activated by proteolytic cascades (e.g., complement system)
- Examples:
  - Complement proteins (C3a, C5a, C3b)
  - Coagulation factors
  - Kinin system (bradykinin)

Acute Inflammation: (e) Mediators of Inflammation

Mediators are produced only when needed

- Mediators are released only after a stimulus, such as:
  - Microbial products (e.g., bacteria, viruses)
  - Substances from necrotic cells
- This ensures inflammation happens only at the right time and place.
- This prevents unnecessary inflammation when there is no threat.

Most mediators are very short-lived

- They are rapidly broken down, inactivated, or removed.
- This prevents inflammation from running out of control.

Mediators can trigger the release of more mediators

- One mediator can activate others → this can amplify the inflammatory response.
- Example:
  - Complement fragments → stimulate histamine release
  - TNF → stimulates IL-1 and chemokine production

Mediators of Inflammation:
(i) *Vasoactive Amines: Histamine and Serotonin*

Histamine

- One of the earliest mediators released in acute inflammation (because it is pre-formed in granules).
- Major source: Mast cells (also in basophils & platelets).
- Main effects:
  - Vasodilation of arterioles → redness, heat
  - Increased venular permeability → oedema (exudate formation)
- Key role:
  - Principal mediator of the immediate transient phase of vascular leakage, by creating interendothelial gaps in venules.
- Triggers for Histamine Release
  - Physical injury: trauma, heat, cold
  - Allergic reactions: antigen binding to IgE on mast cells
  - Complement fragments: C3a, C5a (anaphylatoxins)
  - Neuropeptides: e.g., substance P
  - Cytokines: IL-1, IL-8
- How Histamine Acts
  - Binds to H1 receptors on microvascular endothelial cells.
- This causes:
  - ✓ endothelial contraction
  - ✓ plasma leakage → oedema
  - ✓ vasodilation
- Clinical relevance
  - Antihistamines work by blocking H1 receptors, preventing vascular effects in conditions like urticaria & allergies.
  - Histamine also contracts some smooth muscle (e.g., bronchial smooth muscle → wheezing in asthma).

Mediators of Inflammation:
(i) *Vasoactive Amines: Histamine and Serotonin*

Serotonin (5-Hydroxytryptamine)

- A preformed vasoactive mediator, mainly in platelets.
- Released when platelets aggregate (e.g., during clot formation).
- Causes vasoconstriction but has a minor role compared to histamine.
- Also produced by neuroendocrine cells in the GIT.
- Functions predominantly as a neurotransmitter in the GI tract and CNS.
- Only limited involvement in acute inflammation.

Mediators of Inflammation:

(ii) Arachidonic Acid Metabolites : Prostaglandins & Leukotrienes

- AA Metabolism → 3 Major Product Groups
- AA is processed through two major enzyme pathways, producing three categories of mediators.

1. Cyclooxygenase (COX) Pathway → Prostaglandins & Thromboxane

- **Enzymes:**
  - COX-1 (constitutive; housekeeping)
  - COX-2 (induced during inflammation)
- **Products**
  - Prostaglandins
  - PGE_2 → fever, pain, vasodilation
  - PGD_2 & $PGF_2\alpha$ → vasodilation, ↑ permeability
  - PGI_2 (prostacyclin) → vasodilation, inhibits platelets
  - Thromboxane A_2 (TxA_2)
  - From platelets
  - Vasoconstriction + platelet aggregation

(ii) Arachidonic Acid Metabolites: Prostaglandins

2. Lipoxygenase (LOX) Pathway → (a) Leukotrienes & (b) Lipoxins

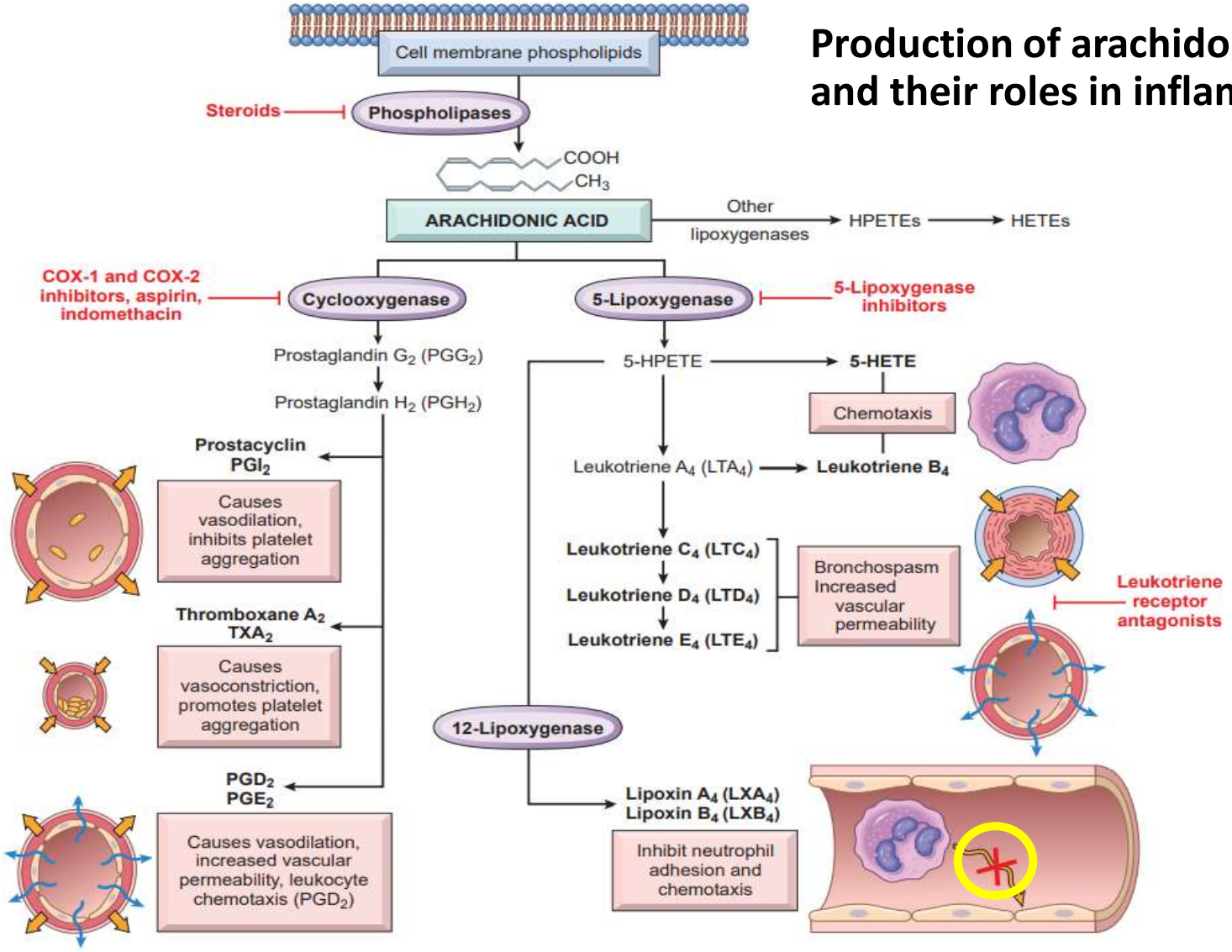
(a) Leukotrienes (Pro-inflammatory)

- Produced by neutrophils, macrophages, mast cells.
  - LTB_4 → strong neutrophil chemoattractant
  - LTC_4, LTD_4, LTE_4 →
    - Bronchoconstriction
    - ↑ Vascular permeability
    - Vasoconstriction

(b) Lipoxins (Anti-inflammatory / Pro-resolving)

- Generated by neutrophils & platelets (12/15-LOX pathway).
  - Lipoxin A_4 & B_4
    - Inhibit neutrophil recruitment
    - Promote resolution of inflammation

Production of arachidonic acid metabolites and their roles in inflammation.



Note the enzymatic activities whose inhibition through pharmacologic intervention blocks major pathways (denoted with a red X).

COX-1, COX-2, Cyclooxygenase 1 and 2; HETE, hydroxyeicosatetraenoic acid; HPETE, hydroperoxyeicosatetraenoic acid.

Pharmacologic Inhibitors of Prostaglandins and Leukotrienes

- **Eicosanoids (prostaglandins, leukotrienes, lipoxins)** play key roles in vascular changes, pain, fever, and leukocyte recruitment. Because of this, many anti-inflammatory drugs work by **blocking their synthesis or actions**.

1. Cyclooxygenase (COX) Inhibitors – NSAIDs

- **Examples:** Aspirin, Ibuprofen, Naproxen
- **Mechanism:** Block **COX-1 & COX-2 → ↓ prostaglandins**
- **Effects:**
- ↓ Pain, ↓ Fever, ↓ Inflammation
- Aspirin irreversibly inhibits COX
- **Note:** NSAIDs **do not** reduce leukotrienes (LOX pathway unaffected)

2. 5-Lipoxygenase (5-LOX) Inhibitors

- **Example:** Zileuton
- **Mechanism:** Blocks leukotriene synthesis
- **Use: Asthma** (because leukotrienes cause bronchoconstriction + edema)

3. Leukotriene Receptor Antagonists

- **Examples:** Montelukast, Zafirlukast
- **Mechanism:** Block leukotriene receptors (especially LTC_4/LTD_4)
- **Use:** Asthma; Allergic rhinitis

4. Corticosteroids (Broad Anti-inflammatory Agents)

- Examples: Prednisolone, Dexamethasone
- Mechanism:
  - ↓ Phospholipase A_2 → ↓ arachidonic acid release
  - ↓ COX-2 expression
  - ↓ Pro-inflammatory cytokines (IL-1, TNF)
  - ↓ iNOS and many inflammation-related genes
- Effect: Very powerful suppression of inflammation
  Use**:** Autoimmune diseases, severe asthma, allergic reactions

5. Dietary Lipids (Fish Oil – Omega-3 Fatty Acids)

- Mechanism:
  - Omega-3 fatty acids are poor substrates for COX & LOX → less inflammatory eicosanoids
  - Promote anti-inflammatory mediators (e.g., resolvins)
- Use: Adjunct in chronic inflammatory states

Mediators of Inflammation: (iii) Cytokines and Chemokines

- Cytokines are small protein messengers produced by many cells that mediate and regulate immune and inflammatory responses.

- **Who Produces Cytokines?**
  - Main producers:
    - Activated lymphocytes
    - Macrophages
    - Dendritic cells
  - Other contributors:
    - Endothelial cells
    - Epithelial cells
    - Fibroblasts / connective tissue cells

- **What Do Cytokines Do?**
  - Coordinate inflammation
  - Recruit and activate leukocytes
  - Regulate the immune response
  - Influence healing and tissue repair

(iii) Cytokines and Chemokines: *Tumour Necrosis Factor and Interleukin-1*

TNF & IL-1 – Key Cytokines

Who produces them?

- Mainly activated macrophages and dendritic cells
- TNF also from T cells and mast cells
- IL-1 also from epithelial cells

What triggers their release?

- Microbial products (via TLRs)
- Dead cells
- Immune complexes
- Foreign bodies
- Physical injury
- IL-1 activation requires the inflammasome

Major Actions of TNF & IL-1

1. Endothelial Activation (most important function)

- They act on endothelial cells to:
  - ↑ Adhesion molecules → E-selectin, P-selectin, ICAM-1, VCAM-1
  - ↑ Chemokines & cytokines
  - ↑ Eicosanoids (PGs, LTs)
  - ↑ Procoagulant activity
- Outcome: promotes leukocyte adhesion → migration → inflammation.

2. Activation of Leukocytes & Local Cells

- TNF increases neutrophil responses and activates macrophages (↑NO production)
- IL-1 stimulates fibroblast collagen synthesis
- IL-1 promotes Th17 responses, strengthening acute inflammation

(iii) Cytokines and Chemokines: *Tumour Necrosis Factor and Interleukin-1*

3. Systemic Acute-Phase Responses

- TNF, IL-1, and IL-6 mediate:
- Fever
- Acute-phase protein production
- Leukocytosis
- TNF also affects metabolism:
- ↑ Lipid & protein breakdown
- ↓ Appetite
- Chronic TNF → cachexia
  (seen in chronic infections, cancers)

Clinical Correlations

- **TNF Antagonists**
  - Very effective for:
    ✓ Rheumatoid arthritis
    ✓ Psoriasis
    ✓ Inflammatory bowel disease
  - But ↑ risk of **tuberculosis** & intracellular infections *(macrophages rely on TNF to kill intracellular microbes)*
- **IL-1 Antagonists**
  - Less effective than TNF blockers (exact reason unclear)
- **Sepsis**
  - Blocking TNF/IL-1 does not improve survival in sepsis (because many other cytokines are involved)

(iii) Cytokines and Chemokines: *Chemokines*

What are chemokines?

- Small cytokines that direct leukocyte movement (chemoattractants).
- Tell leukocytes where to go during inflammation.
- Classified by the pattern of cysteine (C) residues.

- **Four Major Classes**

1. CXC (α-chemokines)

- *Example:* IL-8
- Produced by macrophages & endothelial cells
- Acts mainly on neutrophils → activation + chemotaxis

2. CC (β-chemokines)

- *Examples:* MCP-1, eotaxin, MIP-α, RANTES
- Attract monocytes, eosinophils, basophils, lymphocytes
- Do NOT attract neutrophils

3. C chemokines (γ-chemokines)

- *Example:* Lymphotactin
- Attracts lymphocytes

4. CX3C chemokines

- *Example*: Fractalkine
- Exists in membrane-bound and soluble forms
- Attracts monocytes and T cells, mediates adhesion

(iii) Cytokines and Chemokines: *Chemokines*

Functions of Chemokines

1. Acute Inflammation

- Produced in response to microbes, necrotic cells, and cytokines
- Key actions:
    - ✓ Increase integrin affinity on leukocytes → stronger adhesion
    - ✓ Guide leukocytes via chemotaxis → migrate to infection/injury site

2. Maintenance of Tissue Architecture

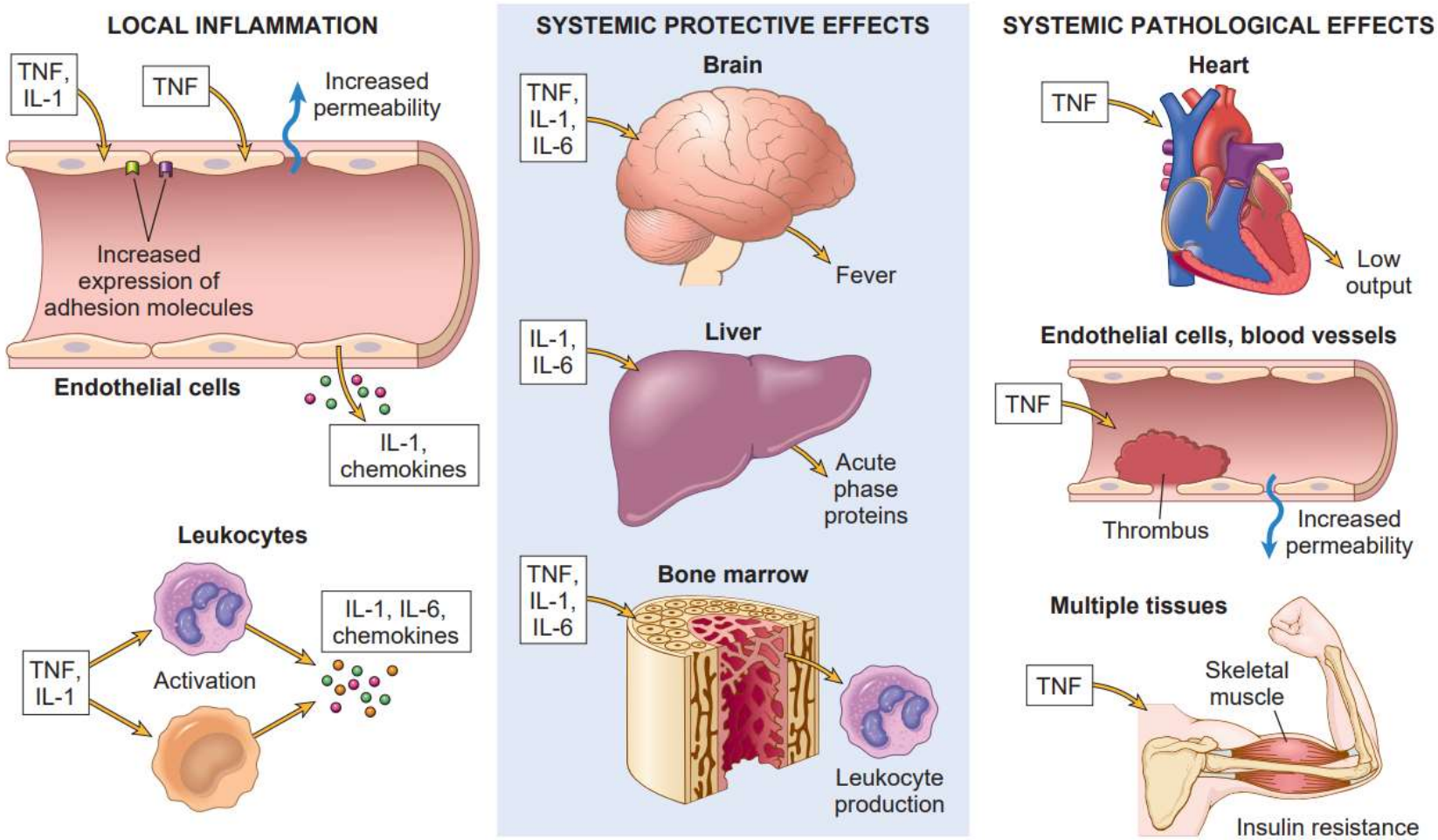
- "Homeostatic chemokines" organise cells in tissues
    - ✓ Arrange T cells & B cells in lymph nodes and spleen
    - ✓ Maintain normal immune structure

Cytokines in Inflammation

Cytokine	Principal Sources	Principal Actions in Inflammation
In Acute Inflammation		
TNF	Macrophages, mast cells, T lymphocytes	Stimulates expression of endothelial adhesion molecules and secretion of other cytokines; systemic effects
IL-1	Macrophages, endothelial cells, some epithelial cells	Similar to TNF; greater role in fever
IL-6	Macrophages, other cells	Systemic effects (acute phase response)
Chemokines	Macrophages, endothelial cells, T lymphocytes, mast cells, other cell types	Recruitment of leukocytes to sites of inflammation; migration of cells in normal tissues
IL-17	T lymphocytes	Recruitment of neutrophils and monocytes
In Chronic Inflammation		
IL-12	Dendritic cells, macrophages	Increased production of IFN-γ
IFN-γ	T lymphocytes, NK cells	Activation of macrophages (increased ability to kill microbes and tumor cells)
IL-17	T lymphocytes	Recruitment of neutrophils and monocytes

IFN-γ, Interferon-γ; IL, interleukin; NK, natural killer; TNF, tumor necrosis factor.
The most important cytokines involved in inflammatory reactions are listed. Many other cytokines may play lesser roles in inflammation. There is also considerable overlap between the cytokines involved in acute and chronic inflammation. Specifically, all the cytokines listed under acute inflammation may also contribute to chronic inflammatory reactions.

Major roles of cytokines in acute inflammation.



Acute Inflammation: (iv) Complement System

Overview

- Complement = plasma proteins that help defend against microbes and contribute to inflammation.
- Produced mainly in the liver, circulate as inactive precursors.
- Activated → form enzymatic cascade → strong amplification.

Activation Pathways

(All generate C3 convertase → cleaves C3)

1. Classical pathway

- *Trigger*: Antigen–antibody complexes (IgM, IgG).
- C1 binds antibody → cascade starts.

2. Alternative pathway

- *Trigger*: Microbial surface molecules (LPS, endotoxin).
- Does NOT require antibody.

3. Lectin pathway

- *Trigger*: Mannose-binding lectin (MBL) binding to microbial carbohydrates.
- Activates C1-like proteins.

Acute Inflammation: (iv) Complement System

Key Step: C3 Cleavage

- C3 convertase splits C3 → C3a + C3b
  - C3a = inflammatory mediator
  - C3b = binds microbe → opsonin
- C3b helps form C5 convertase → cleaves C5 → C5a (inflammatory) + C5b (starts membrane attack complex formation)

Main Functions of Complement

1. Inflammation

- C3a, C4a, C5a = anaphylatoxins → induce mast cell degranulation, ↑ vascular permeability
- C5a = strong chemotactic factor for neutrophils, monocytes, eosinophils
- C5a activates AA metabolism → more inflammatory mediators

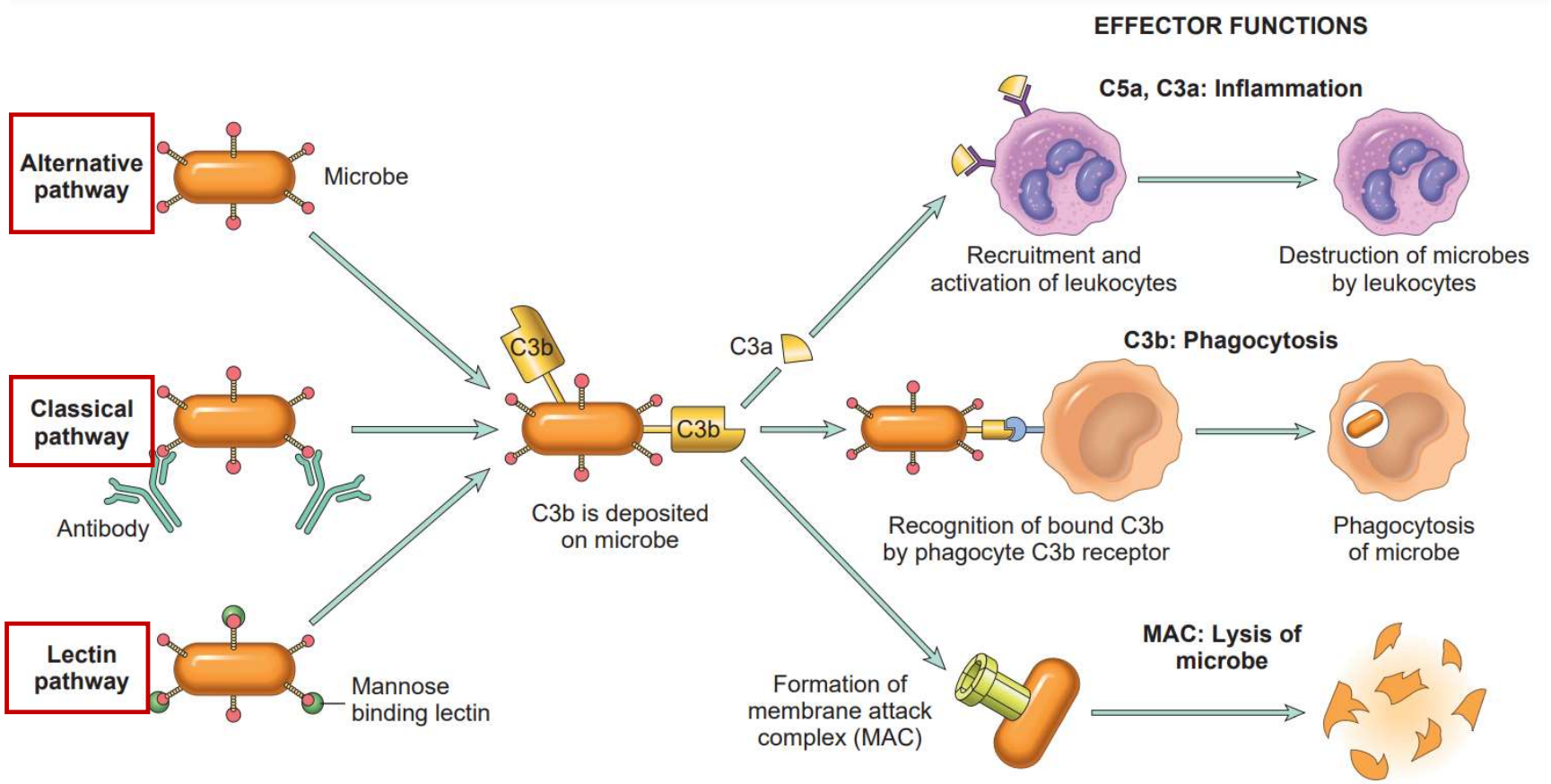
2. Opsonisation

- C3b + iC3b coat microbes → phagocytes (with C3b receptors) bind & engulf more easily

3. Cell lysis

- C5b–C9 = Membrane Attack Complex (MAC) → creates pores → osmotic lysis → especially important for Neisseria (deficiency → recurrent Neisseria infections)

Activation and functions of the complement system.



Activation of complement by different pathways leads to cleavage of C3. The functions of the complement system are mediated by breakdown products of C3 and other complement proteins and by the membrane attack complex (MAC).

Acute Inflammation: (iv) Complement System

Complement Regulation (Prevents Host Damage)

- Normal cells express regulatory proteins to stop excessive activation.

1. C1 inhibitor (C1-INH)

- Blocks classical pathway activation
- Deficiency → hereditary angioedema

2. DAF & CD59 (GPI-anchored)

- DAF: prevents C3 convertase formation
- CD59: prevents MAC assembly
- GPI-anchor deficiency → excessive complement activation →Paroxysmal Nocturnal Hemoglobinuria (PNH)

3. Factor H

- Inhibits alternative pathway
- Promotes breakdown of C3b
- Mutation → atypical hemolytic uremic syndrome (HUS)

Clinical Relevance

- **Excess complement activation**
  - Occurs in autoimmune diseases→ antibodies deposit on host tissue → complement → inflammation & damage

- **Complement deficiencies**
  - Early components (C1, C4) → ↑ immune complex disease
  - C3 deficiency → severe recurrent infections
  - MAC (C5–C9) deficiency → Neisseria infections

Acute Inflammation: (v) Other Mediators of Inflammation

Platelet-Activating Factor (PAF)

- Produced by platelets, basophils, mast cells, neutrophils, macrophages, endothelial cells
- Originally named for causing platelet aggregation
- Powerful inflammatory mediator with multiple **effects**:
  - Vasodilation
  - ↑ Vascular permeability (very potent—may exceed histamine)
  - Bronchoconstriction
  - Leukocyte activation
  - Platelet activation

Products of Coagulation

- Coagulation and inflammation are closely linked.
- Protease-activated receptors (PARs) on many cells → activated by clotting factors (e.g., thrombin).
  - Major role: platelet activation
- Fibrin split products (fibrinopeptides):
  - Promote inflammation
  - Can increase vascular permeability and attract leukocytes
- In tissue injury, clotting and inflammation often occur together → difficult to separate cause vs effect.

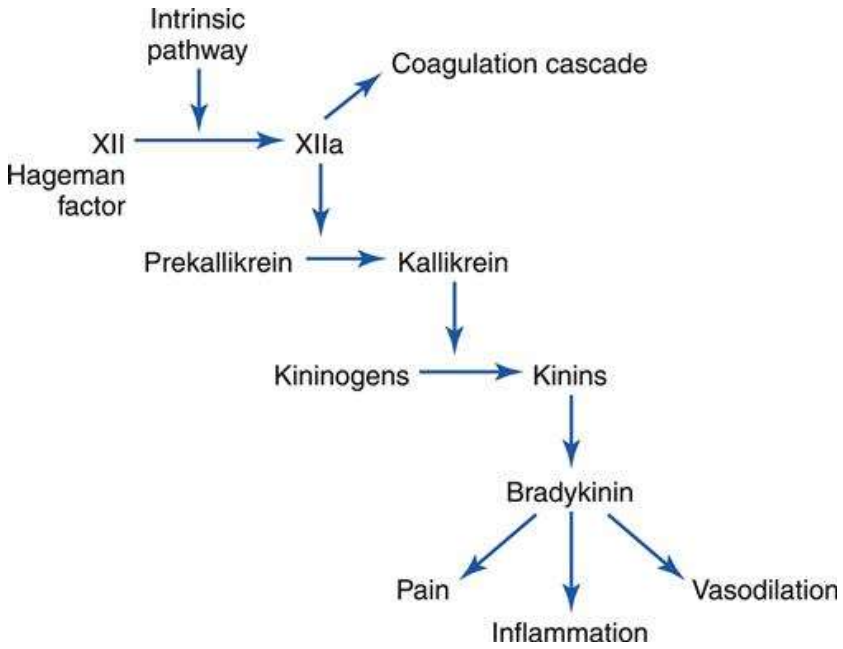
Acute Inflammation: (v) Other Mediators of Inflammation

Kinins

- Kinins are vasoactive peptides produced from plasma proteins (kininogens).
- They are generated by kallikreins, a group of proteases.

Bradykinin

- Formed when kallikrein cleaves high-molecular-weight kininogen.
- ***Major effects*** (similar to histamine):
  - ↑ Vascular permeability
  - Vasodilation
  - Smooth muscle contraction
  - Pain (important mediator of inflammatory pain)
- Short-acting → rapidly broken down by kininase.



Acute Inflammation: (v) Other Mediators of Inflammation

Neuropeptides

What are they?

- Small peptide molecules released by sensory nerves and leukocytes.
- Help initiate and regulate inflammation.

Where are they produced?

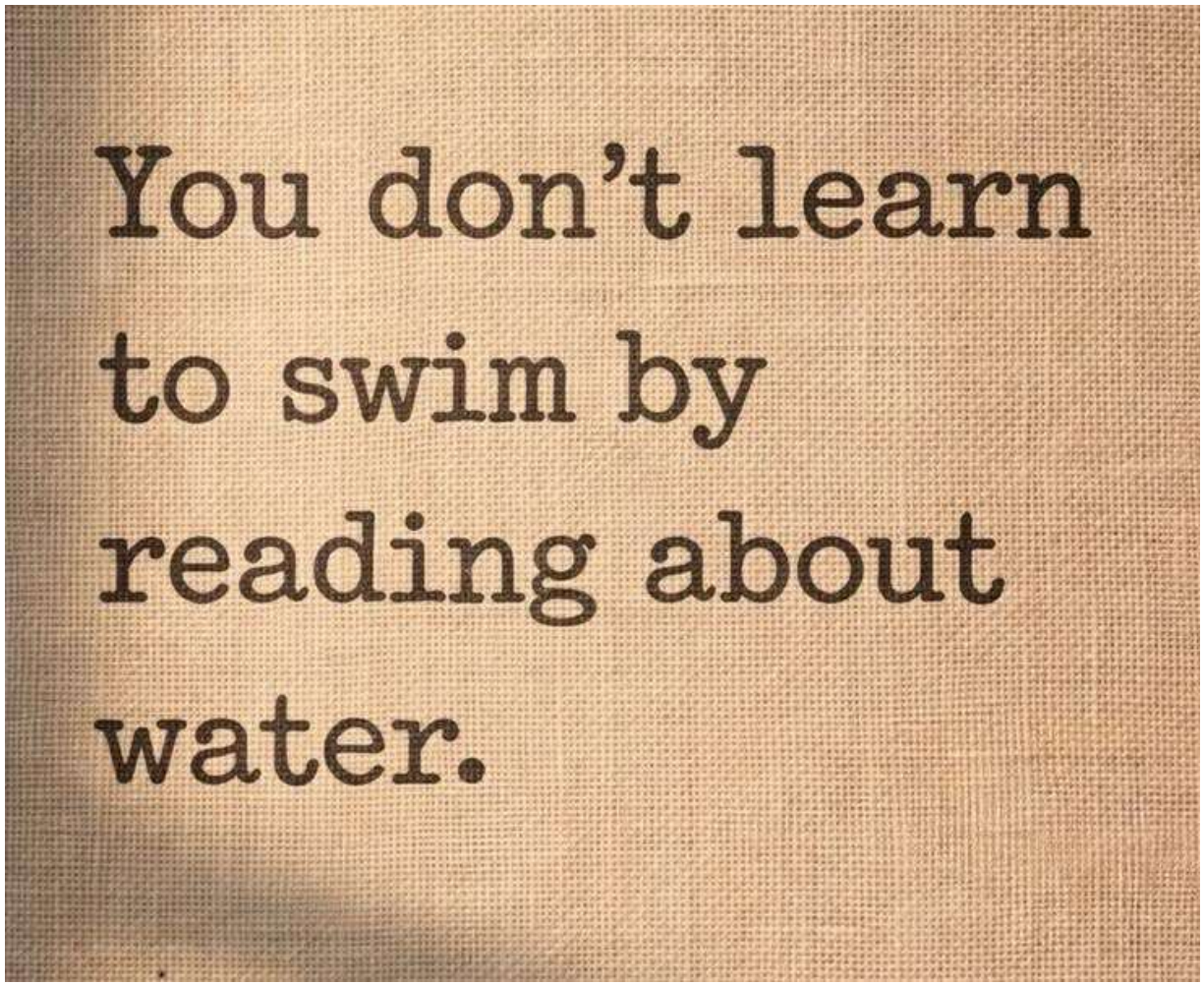
- In both the central and peripheral nervous systems.
- Nerve fibres rich in neuropeptides (e.g. substance P) are abundant in the lungs and gastrointestinal tract.

- **Substance P**
  - Major neuropeptide in inflammation.
  - Transmits pain signals.
  - Increases vascular permeability.
  - Contributes to neurogenic inflammation.

- **Neurokinin A**
  - Works similarly to substance P.
  - Supports inflammatory signalling pathways.

Why neuropeptides matter in inflammation

- Leukocytes have receptors for neuropeptides → allows cross-talk between the nervous and immune systems.
- Neural signals can modulate inflammation.
- Example:
  - Vagus nerve activation reduces the release of pro-inflammatory cytokines (e.g. TNF).
  - This “anti-inflammatory reflex” is being explored in therapy → vagus nerve stimulation trials in rheumatoid arthritis.


You don't learn
to swim by
reading about
water.

(f) Morphologic Patterns of Acute Inflammation

- The morphologic hallmarks of acute inflammation are dilation of small blood vessels and accumulation of leukocytes and fluid in the extravascular tissue.
- Depending on the cause, severity and tissue involved, additional morphologic patterns may appear on top of these basic changes.
- Recognising the morphologic patterns provide valuable clues of underlying cause

(f) Morphologic Patterns of Acute Inflammation

(1) Serous Inflammation

- Serous inflammation involves accumulation of cell-poor, clear fluid in spaces caused by injury or in body cavities (pleura, peritoneum, pericardium).
- The fluid is not rich in cells or microbes; it mainly comes from plasma leakage due to increased vascular permeability or from mesothelial cell secretion.
- A classic example is a skin blister (burn or viral infection), where serous fluid collects beneath the damaged epidermis.



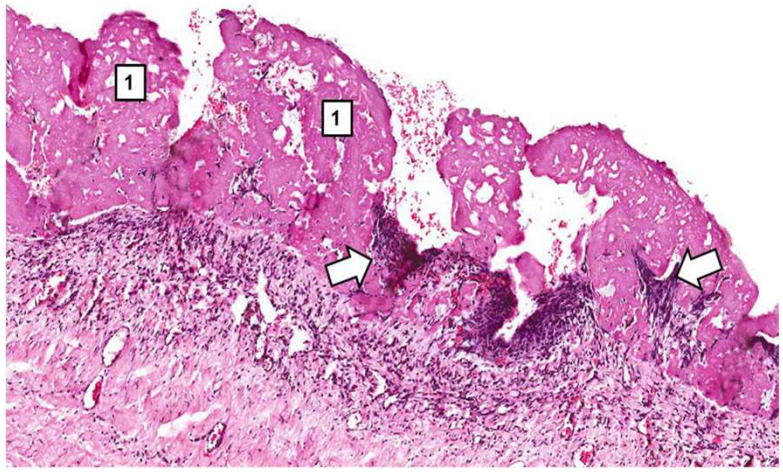
(f) Morphologic Patterns of Acute Inflammation

(2) Fibrinous Inflammation

- Fibrinous inflammation occurs when vascular permeability increases enough for fibrinogen to leak out and form fibrin, especially in body cavity linings (pleura, pericardium, meninges).
- Microscopically, fibrin appears as an eosinophilic meshwork, and the exudate may be removed by fibrinolysis or cleared by macrophages.
- If not removed, fibrin becomes organised into scar tissue, which can cause complications such as fibrous thickening and possible obliteration of the pericardial space.



Deposits of fibrin on the pericardium. Aka bread and butter appearance.



- A pink fibrin meshwork (amorphous eosinophilic material) coats the surface of the pericardium (1).
- Reactive mesothelial cells line the surface (arrows), showing enlargement and prominence.
- Inflammatory cells are present within the underlying pericardial tissue, consistent with fibrinous pericarditis.

(f) Morphologic Patterns of Acute Inflammation

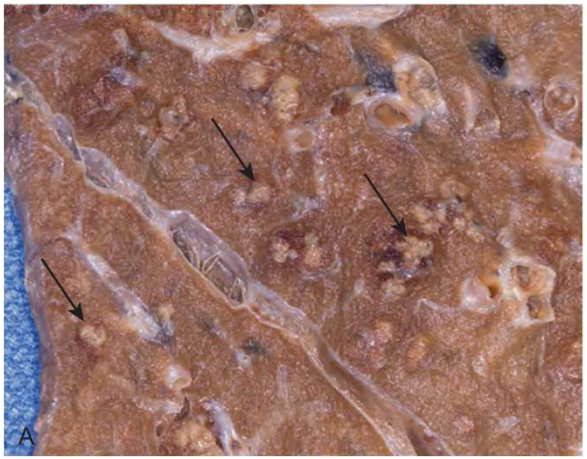
(3) Purulent (Suppurative) Inflammation and Abscess

Purulent / Suppurative inflammation

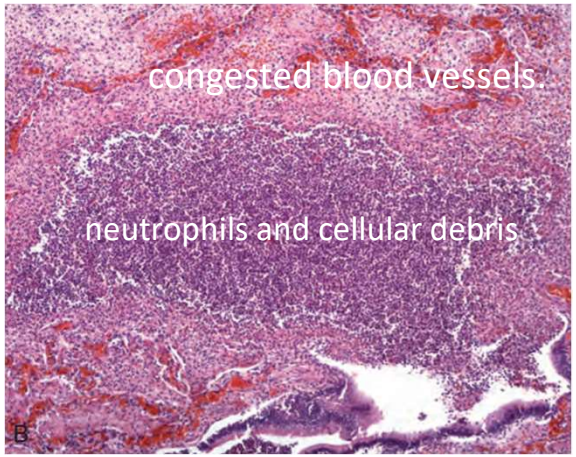
- Defined by pus, which contains neutrophils, necrotic cell debris, and oedema fluid.
- Most commonly caused by pyogenic bacteria (e.g., staphylococci) that induce liquefactive necrosis.
- Classic clinical example: acute suppurative appendicitis.

Abscess

- A localised collection of pus within tissue, an organ, or a confined space.
- Caused by deep seeding of pyogenic bacteria.
- Structure of an abscess:
  - Central necrotic area → liquefied dead neutrophils and tissue.
  - Surrounding zone of viable neutrophils.
  - Outer rim with dilated vessels, inflammatory cells, and fibroblasts (early repair).
- Over time, abscesses may become walled off and replaced by scar tissue.



Multiple bacterial abscesses (arrows) in the lung in a case of bronchopneumonia



The abscess contains neutrophils and cellular debris and is surrounded by congested blood vessels.

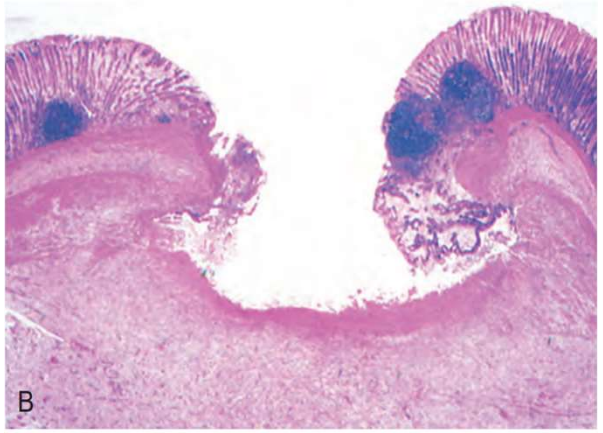
(f) Morphologic Patterns of Acute Inflammation

(4) Ulcers

- An ulcer is a localised loss of surface tissue caused by sloughing of necrotic, inflamed tissue, occurring only on surface-lined structures (skin or mucosa).
- Common sites include the mouth, stomach, intestines, genitourinary tract, and lower limbs, especially in conditions with poor blood supply (e.g., diabetes, peripheral vascular disease, sickle cell disease).
- Acute ulcers show neutrophilic infiltration and vascular dilation at the edges.
- **Chronic ulcers** show fibroblast proliferation, scarring, and infiltration by lymphocytes, macrophages and plasma cells (mixed acute + chronic inflammation).
- Classic example: peptic ulcer of the stomach/duodenum, where acute and chronic inflammation coexist.

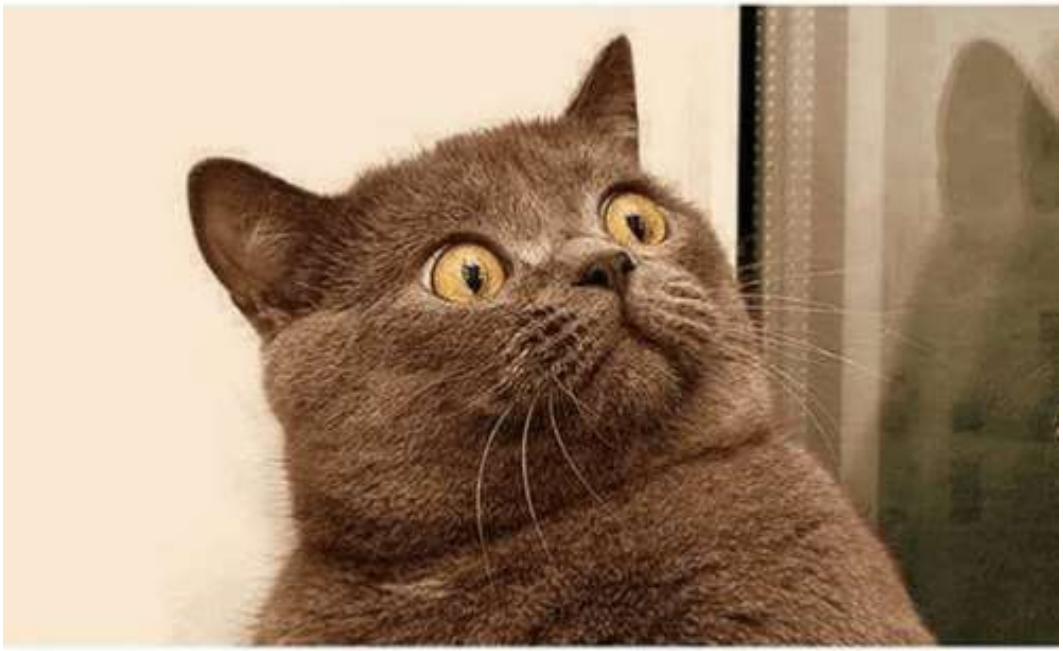


A chronic duodenal ulcer



A duodenal ulcer crater with an acute inflammatory exudate in the base.

My friend's reaction



When I ask for an
extra sheet in the exam

(g) Outcomes of Acute Inflammation

Factors That Modify Inflammation

- Inflammation varies depending on:

1. **Factors related to the organism (injury/infection)**
   - Type, intensity, and site of injury
   - Virulence & load of the infective agent
   - Portal of entry of microbes

2. **Factors related to the host**
   - General health (malnutrition, diabetes, alcoholism ↑ susceptibility)
   - Immune status (immunodeficiency, neutropenia)
   - Local factors: ischemia, necrosis, foreign bodies
   - Type of tissue (some tissues regenerate, some cannot)

(g) Outcomes of Acute Inflammation

Three Possible Outcomes of Acute Inflammation

(1) Complete Resolution

- Occurs when injury is short-lived, mild, with minimal tissue destruction.
- Requires tissues that can regenerate.
- Resolution involves:
  - Removal of dead cells & microbes by macrophages
  - Drainage of oedema via lymphatics
  - Regeneration of tissue → return to normal

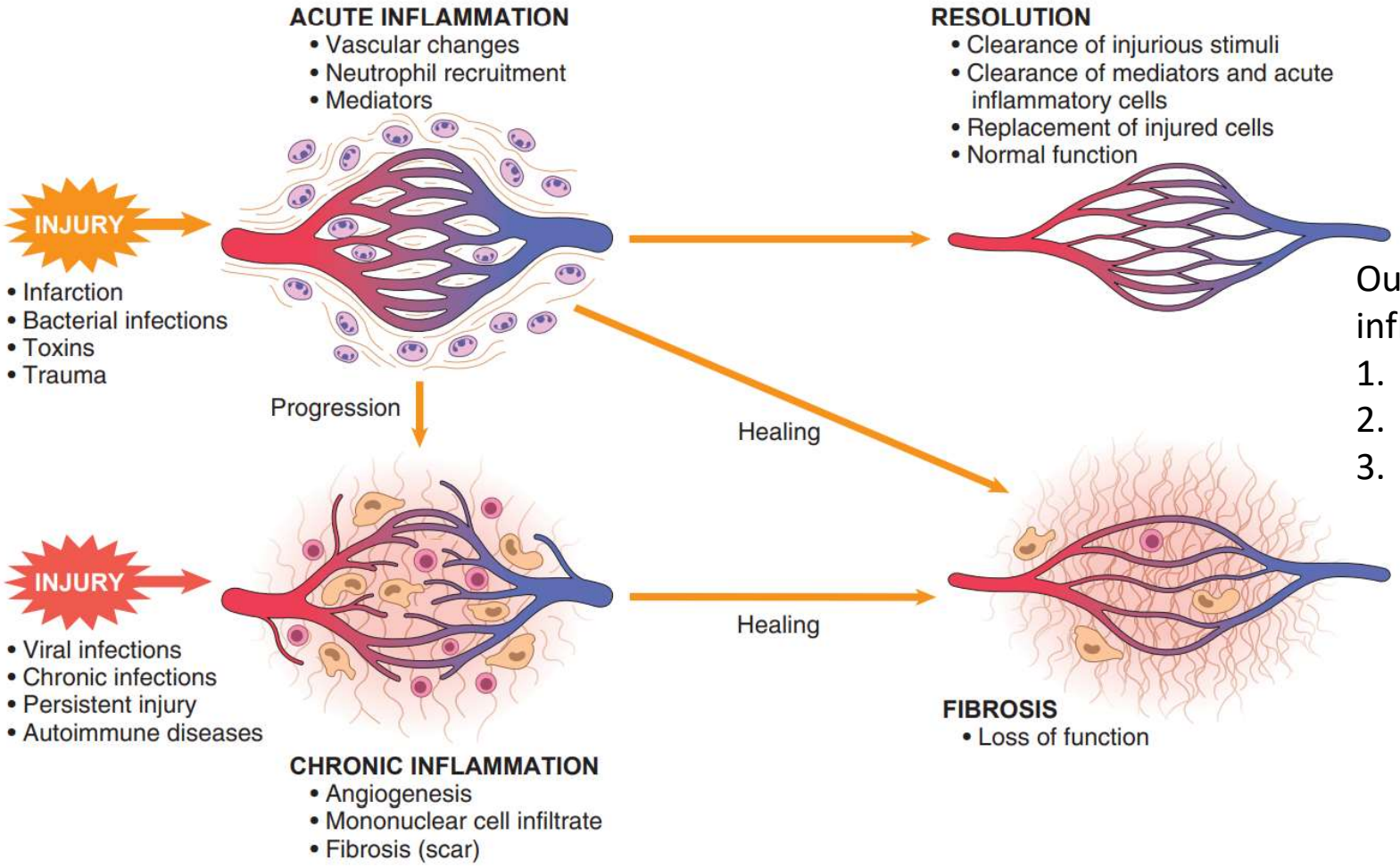
(2) Healing by Fibrosis (Scar Formation)

- Occurs when:
  - Tissue destruction is extensive
  - Tissue cannot regenerate
  - Fibrin exudate cannot be cleared (e.g., in serous cavities → pleura, peritoneum)
  - Fibroblasts grow into the area and form fibrous scar tissue
- This process is called organisation

(3) Progression to Chronic Inflammation

- Happens when acute inflammation cannot be resolved due to:
  - Persistent infection or stimulus
  - Interference with normal healing
  - Leads to chronic inflammation with ongoing tissue injury and repair.

(g) Outcomes of acute inflammation



Outcomes of acute inflammation:

1. resolution,
2. healing by fibrosis,
3. chronic inflammation.

(h) Summary of Acute Inflammation

Acute inflammation is an immediate protective response to microbes, dead cells or injury.

Its aim: eliminate the cause, remove damaged tissue, and initiate repair.

- **Vascular Changes – Cardinal Signs**
  - Vasodilation → ↑ blood flow → Redness (rubor), Heat (calor)
  - ↑ Vascular permeability → protein-rich oedema → Swelling (tumor)
  - Chemical mediators (PGs, bradykinin, neuropeptides) → Pain (dolor)
  - Pain + tissue injury → Loss of function (functio laesa)
- **Cellular Events**
  - Neutrophils arrive first → adhesion, transmigration, chemotaxis
  - Activation by microbes & mediators
  - Phagocytosis: recognition → engulfment → phagolysosome killing
  - Killing via ROS, NO, lysosomal enzymes (may also damage host tissue)
- **Key Mediators**
  - Histamine, serotonin
  - Prostaglandins, leukotrienes, lipoxins (AA pathway)
  - Cytokines (TNF, IL-1, IL-6), chemokines (IL-8)
  - Complement (C3a, C5a, C3b, MAC)
  - PAF, bradykinin, neuropeptides
- **Outcomes**
  - Resolution – minimal damage, tissue regenerates
  - Scar formation – severe injury / non-regenerative tissue
  - Chronic inflammation – persistent stimuli

Reference

- Kumar, V., Abbas, A. K., Aster, J. C., Turner, J. R., & Perkins, J. A. (2021). *Robbins & Cotran pathologic basis of disease*. 10th Edn. Elsevier.



My friends when I’m still writing the answers in Exam